Tea Blending Optimisation: A Comprehensive Analysis

STA5074Z Prescriptive Analytics

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Contents

1 Introduction

Blending problems are a classic challenge in operations research, arising in diverse fields such as food processing, metallurgy, and energy. In the tea industry, blending is crucial to achieve consistent product quality, balance raw material costs, and adapt to fluctuating supplies. Every cup of tea on the supermarket shelf is the result of a carefully orchestrated process. Different grades and types of tea leaves from various origins are combined to meet precise targets for taste, colour, aroma and other quality parameters, all while minimising production costs and ensuring supply constraints are not violated.

In the context of tea manufacturing, the blending optimisation problem involves selecting the optimal composition of raw teas to satisfy desired blend qualities for multiple final blends. This not only impacts the sensory profile and brand identity of each product, but also has substantial operational and economic implications. By applying advanced computational techniques, such as mathematical programming and metaheuristics, producers can automate and enhance the blending decision-making process, yielding better margins, consistent quality and greater responsiveness to market demands.

Below, we formally define the tea blending optimisation problem and systematically explore several approaches for solving it. The problem requires determining the least-cost combination of raw materials to produce final blends that meet specific quality characteristics, as formalised by Fomeni[1].

The primary objectives are:

1. To implement and solve a cost-minimisation model using Linear Programming (LP).
2. To solve the same problem using Simulated Annealing (SA) metaheuristic and compare performance.
3. To formulate and solve a multi-objective model using Chebyshev Goal Programming.
4. To explore Genetic Algorithms (GAs) as an alternative solution approach.

2 Problem Formulation and Data Preparation

2.1 Notations and Parameters

- **Indices**:
  - i: Index for raw materials
  - j: Index for final blends
  - k: Index for quality characteristics
- **Parameters**:
  - P_i: Unit cost of raw material i (£/Kg)
  - D_j: Weekly demand for final blend j (Kgs)
  - a_i: Weekly availability of raw material i (Kgs)
  - g_{ik}: Grade of characteristic k in one unit of raw material i
  - s_{jk}: Required target score of characteristic k in one unit of final blend j
- **Decision Variables**:
  - x_{ij}: Amount (in Kgs) of raw material i to be used in final blend j

2.2 Data Source

This project uses the real-world data from Fomeni[1], extracted from tables 2 and 3 of the paper. These data represent actual tea blending scenarios from a UK-based tea company. The data are loaded and prepared in section 3 below.

3 Data Implementation (Fomeni 2018)

This section implements the tea blending optimisation problem using the actual data provided in Fomeni[1]. The data consist of 60 raw materials and 10 final blends, each with 6 quality characteristics.

3.1 Data Loading

```
## === Data Summary ===
## Number of raw materials: 60
## Number of blends: 10
## Number of quality characteristics: 6
## Price range: £0.0120 - £0.0470 per kg
## === Raw Materials Data (Fomeni Table 2 - First 10 rows) ===
```

Table 1: Raw Materials: Costs (£/kg), Availability (kg), and Quality Characteristics (First 10)

Material_ID	Cost	Availability	C1	C2	C3	C4	C5	C6
R1	0.0240	5,000	5	4	9	6	5	5
R2	0.0167	1,000	5	3	9	4	5	1
R3	0.0235	25,000	5	4	9	6	5	1
R4	0.0170	1,000	5	3	9	4	5	1
R5	0.0230	25,000	5	4	9	6	5	1
R6	0.0265	20,000	7	4	9	6	7	1
R7	0.0265	20,000	7	4	9	6	7	1
R8	0.0294	1,000	8	4	9	6	8	5
R9	0.0284	15,000	8	4	9	6	8	1
R10	0.0284	15,000	8	4	9	6	8	1

```
## === Blends Data (Fomeni Table 3) ===
```

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B1	100,000	2.3	3.0	3.0	2.0	2.32	1
B2	100,000	3.8	4.7	2.9	3.0	3.80	1
B3	60,000	6.6	5.2	6.1	6.9	6.60	1
B4	100,000	6.2	5.2	5.3	6.4	6.20	1
B5	100,000	4.7	5.1	4.5	5.0	4.70	1
B6	50,000	8.4	4.7	6.7	7.3	8.40	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B7	10,000	4.0	0.5	1.0	5.0	4.00	1
B8	5,000	5.9	0.5	1.0	3.9	5.90	1
B9	10,000	7.0	3.9	9.0	5.9	7.00	1
B10	20,000	4.0	2.0	3.0	2.0	4.00	9

```
## === Feasibility Check ===
## Total availability: 830500 kg
## Total demand: 555000 kg
## Surplus availability: 275500 kg (49.64%)
```

3.2 Data Characteristics

```
## === Summary Statistics ===
## Raw Material Prices (£/kg):
```

Table 3: Price Distribution Statistics

Statistic	Value
Mean	£0.0217
Median	£0.0195
Minimum	£0.0120
Maximum	£0.0470
Standard Deviation	£0.0076

```
## Raw Material Availability:
```

Table 4: Availability Distribution Statistics (kg)

Statistic	Value_kg
Mean	13842
Median	5000
Minimum	500
Maximum	80000
Total	830500

```
## Quality Characteristics:
```

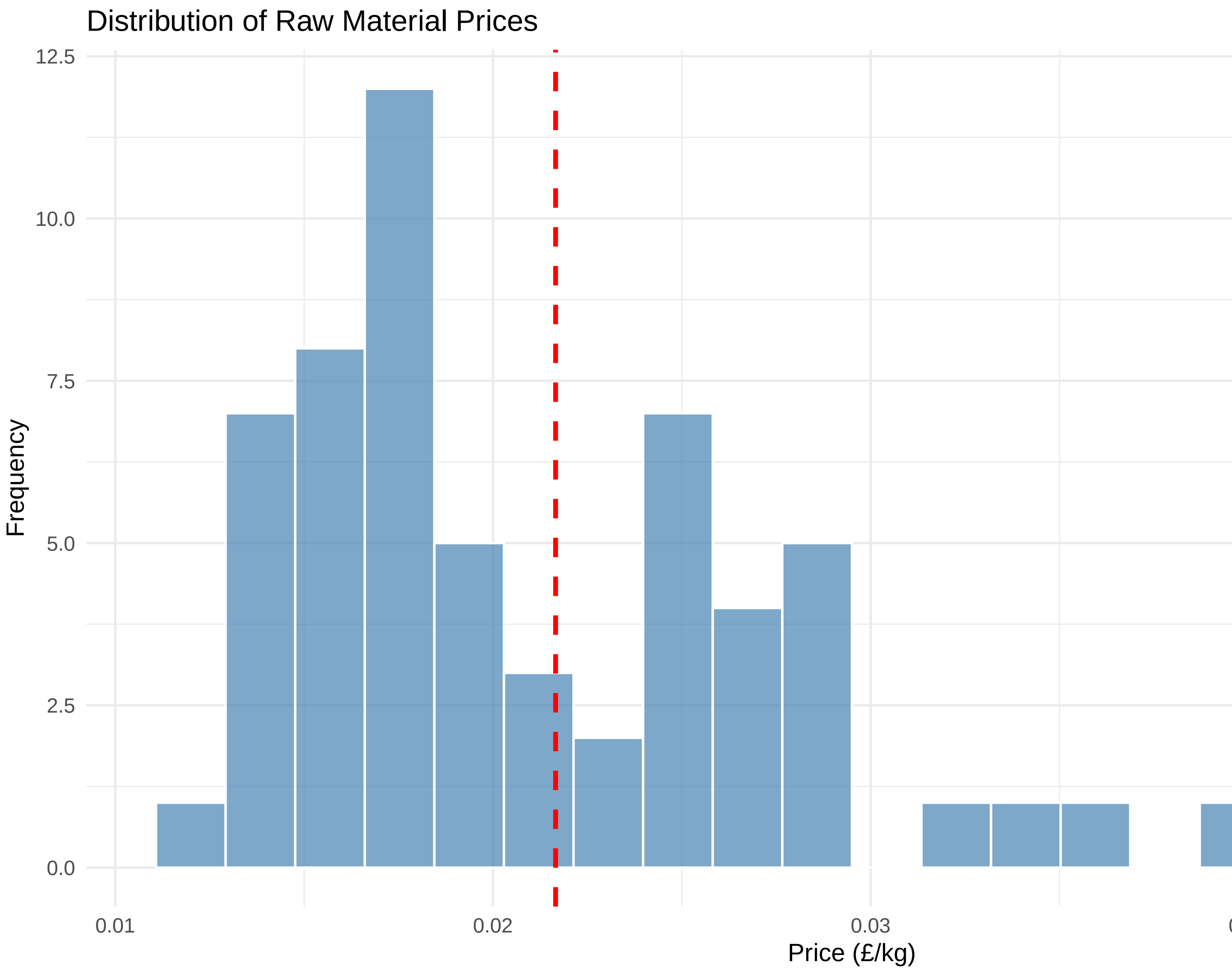
Table 5: Quality Characteristics Summary Statistics

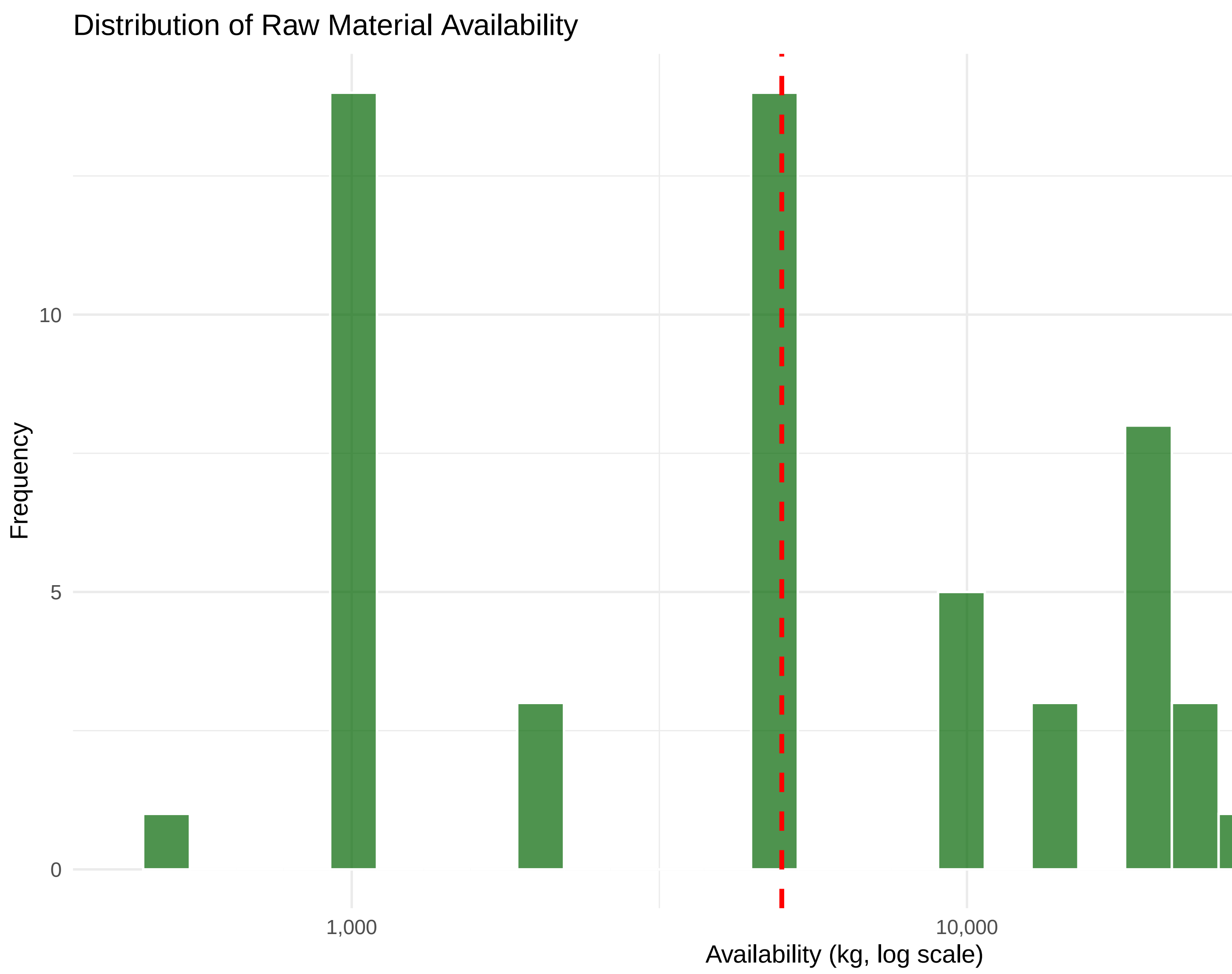
Characteristic	Minimum	Maximum	Mean	Median	Std_Dev
C1	2.0	9	5.500000	5.0	1.917802
C2	0.5	8	3.333333	3.0	1.990798
C3	1.0	9	4.316667	4.0	2.507559
C4	2.0	8	4.500000	3.5	2.452947
C5	2.0	9	5.500000	5.0	1.917802
C6	1.0	9	1.933333	1.0	1.858254

```
## Blend Demands:
```

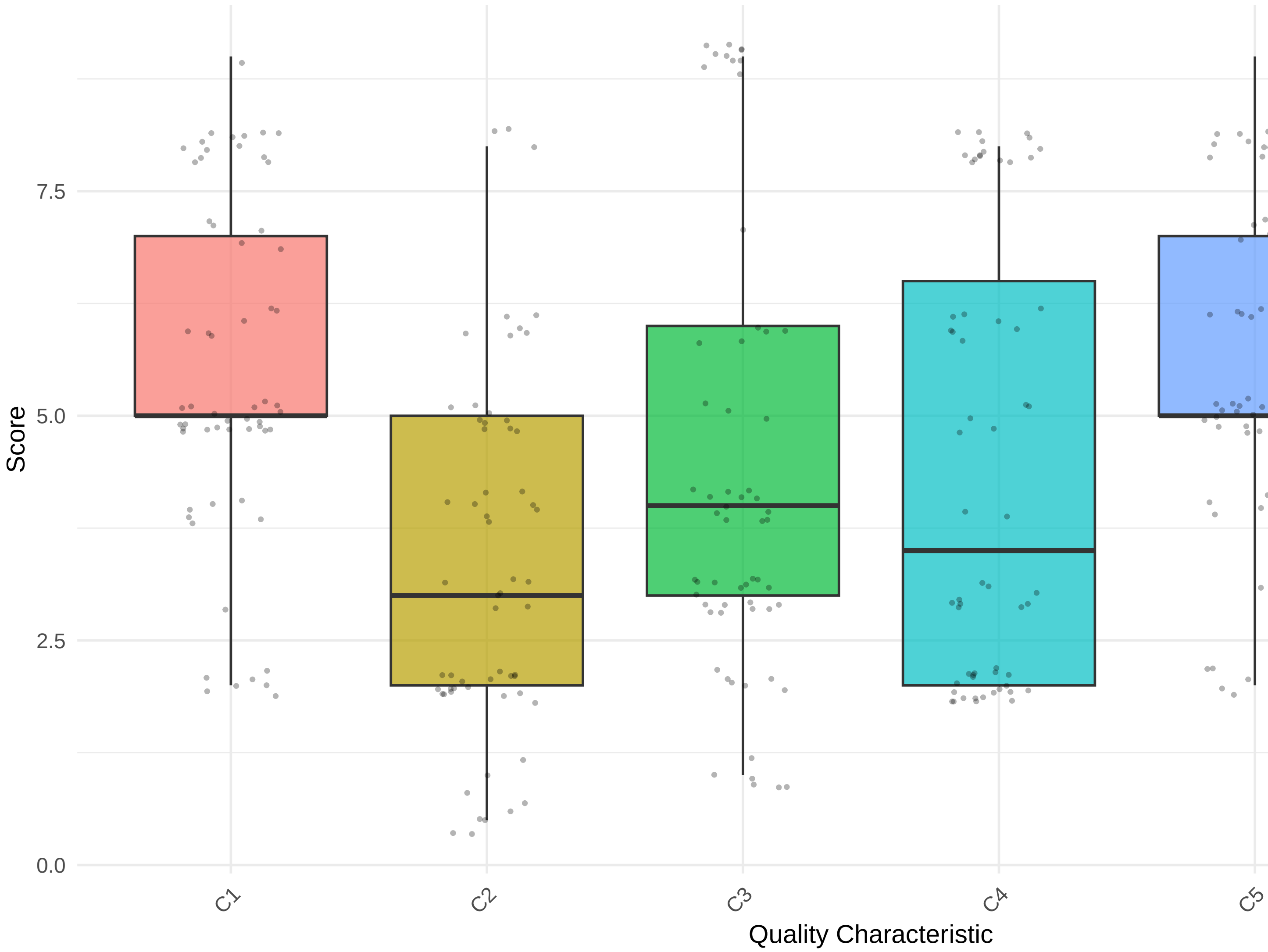
Table 6: Blend Demand Statistics (kg)

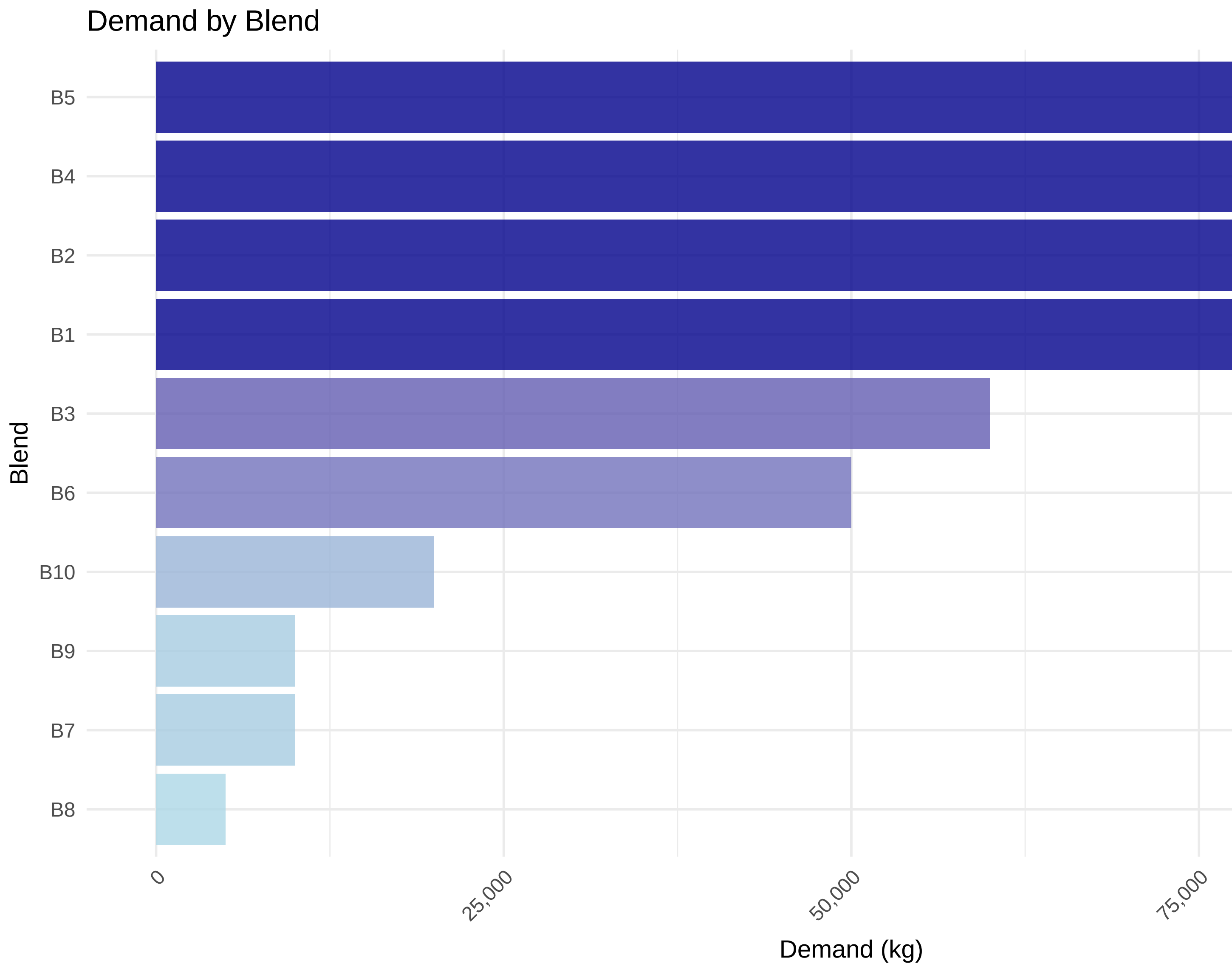
Statistic	Value_kg
Mean	55500
Median	55000
Minimum	5000
Maximum	100000
Total	555000


Distribution of Raw Material Prices
12.5
10.0
7.5
5.0
2.5
0.0
Frequency
0.01
0.02
0.03
Price (£/kg)

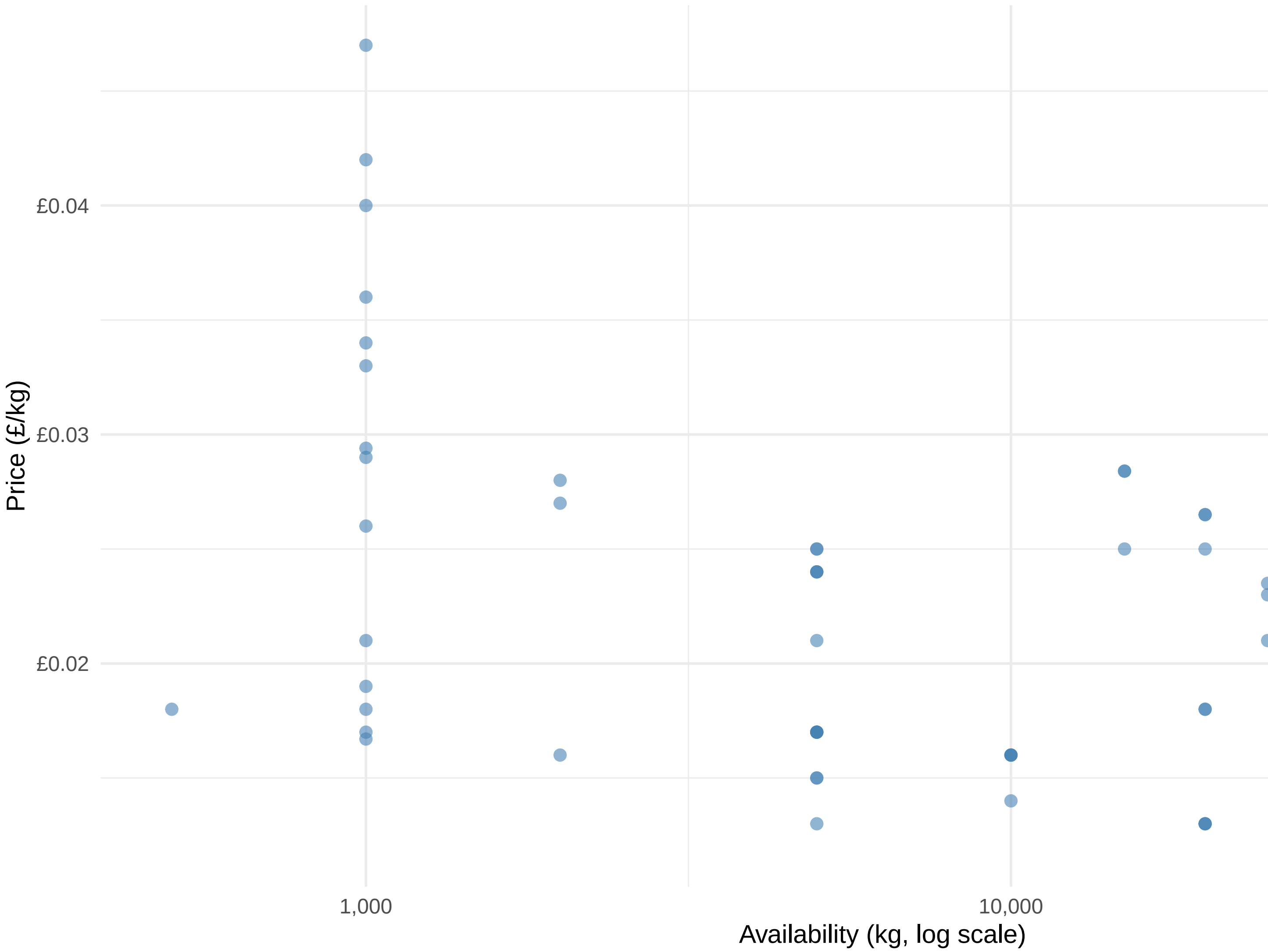

Distribution of Raw Material Availability
Frequency
10
5
0
1,000
10,000
Availability (kg, log scale)

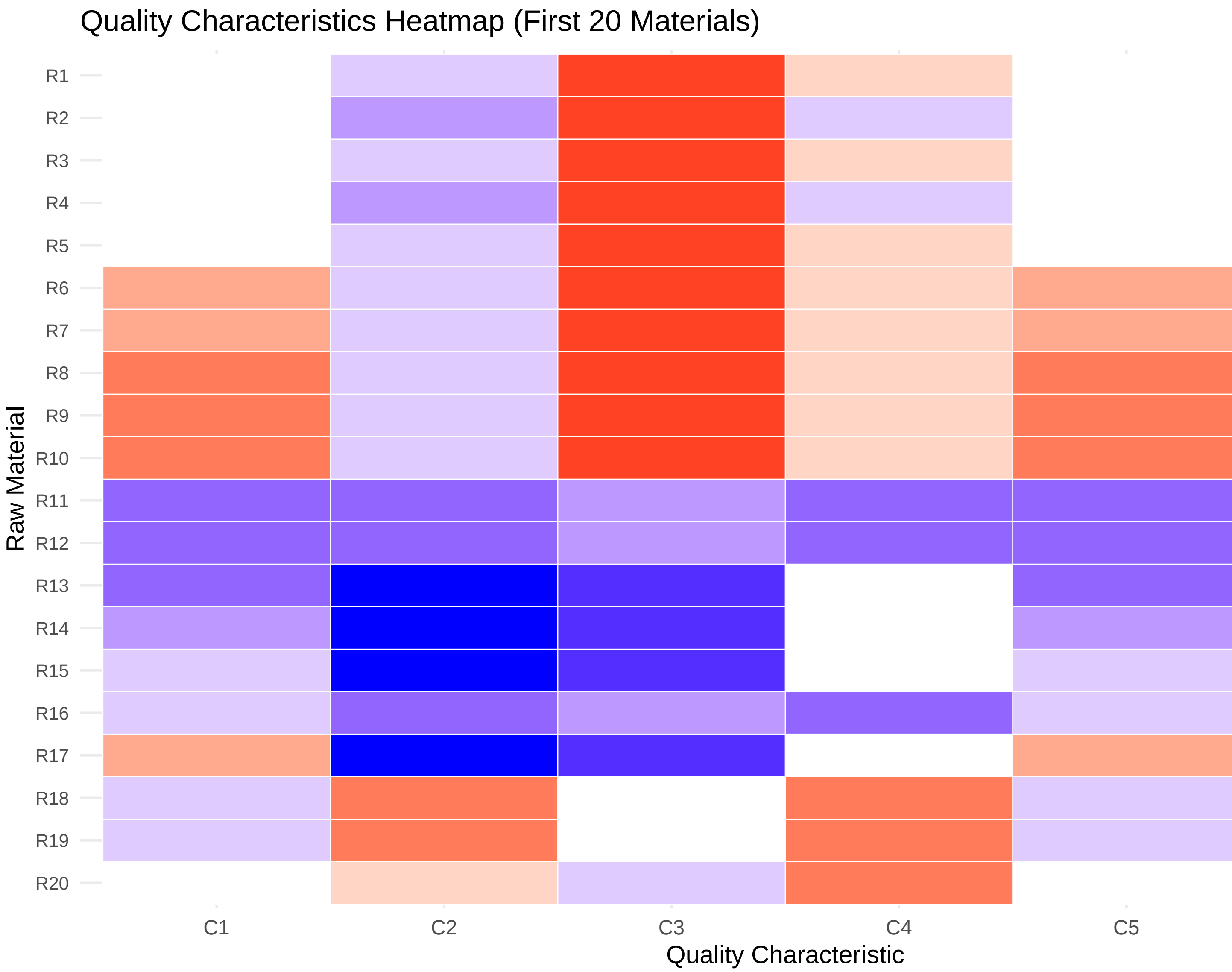
Distribution of Quality Characteristics Across Raw Materials




Demand by Blend
Blend
B5
B4
B2
B1
B3
B6
B10
B9
B7
B8
0
25,000
50,000
75,000
Demand (kg)

Price vs Availability of Raw Materials





4 Approach 1: Cost Minimisation using Linear Programming

This section implements the cost-minimisation model using Linear Programming with parametric relaxation of the characteristic-matching constraints. This involves implementing a cost-minimisation LP model that finds the optimal combination of raw materials to produce blends meeting demand and quality constraints. This implementation uses Rglpk which provides a straightforward matrix-based formulation that is well-suited for this problem type.

4.1 Model Formulation

Objective Function: Minimise total cost

$$\min \sum_i \sum_j P_i x_{ij}$$

Subject to Constraints:

1. Raw Material Availability: $\sum_j x_{ij} \leq a_i \quad \forall i$

2. Blend Demand: $\sum_i x_{ij} \geq D_j \quad \forall j$
3. Characteristic Score Relaxation:
   - $\sum_i (g_{ik} - (s_{jk} + \epsilon_{jk}))x_{ij} \leq 0 \quad \forall j, k$
   - $\sum_i (g_{ik} - (s_{jk} - \epsilon_{jk}))x_{ij} \geq 0 \quad \forall j, k$
4. Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

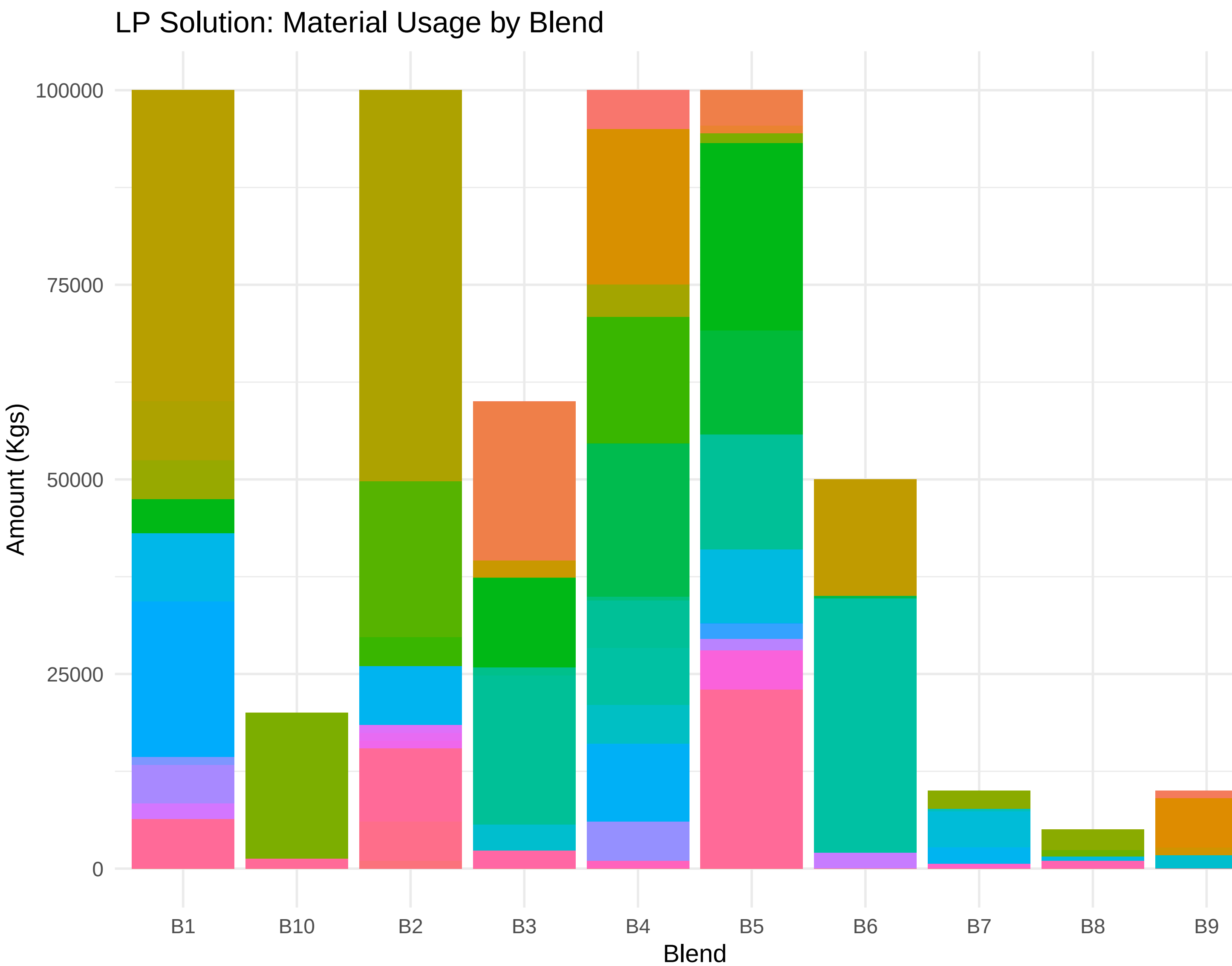
4.2 Implementation

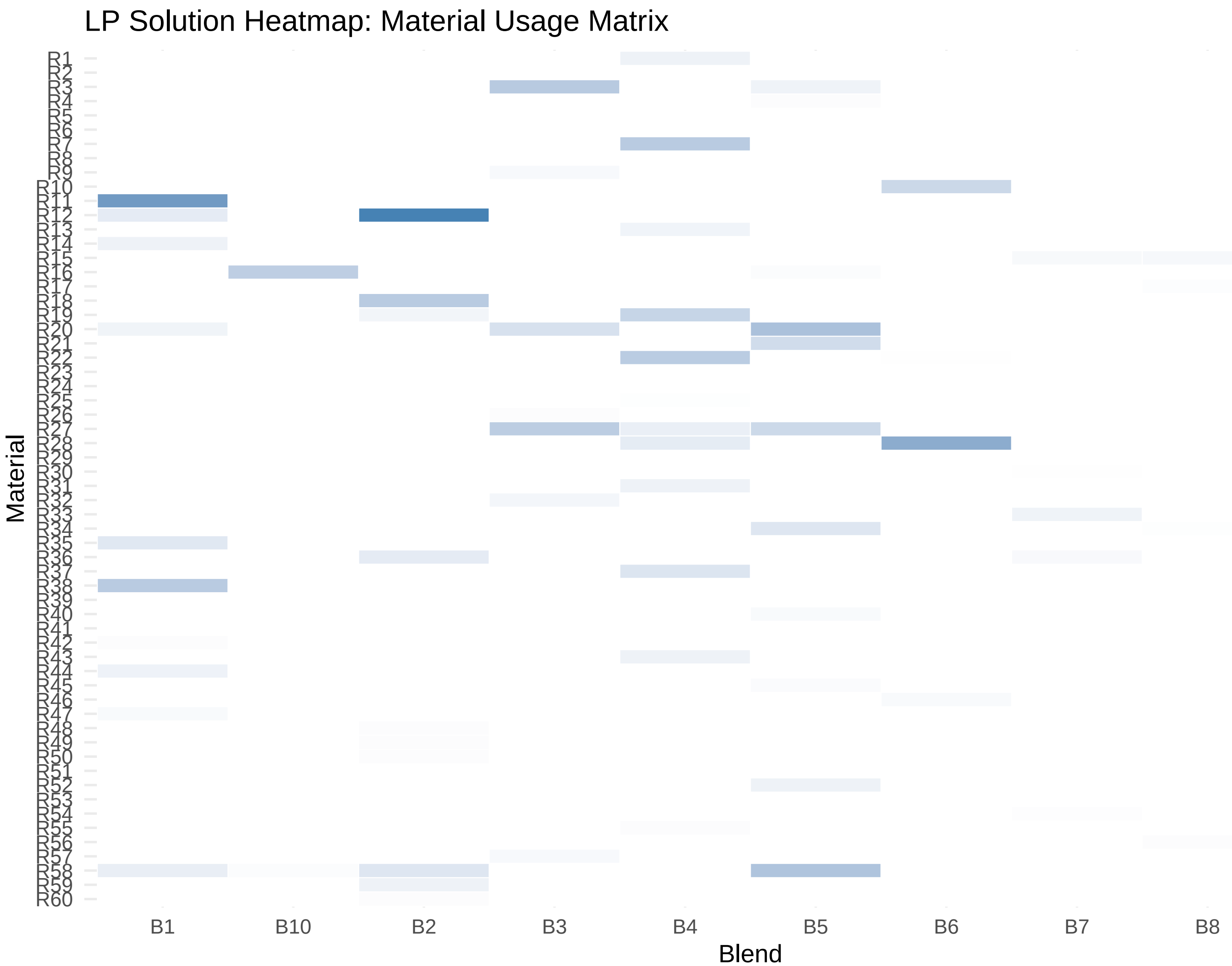
```
## === LP Solution Results ===

## Solver Status: 0 (0 = optimal)

## Optimal Total Cost: £9087.69

## Total material used: 555000.00 Kgs

## Total demand required: 555000.00 Kgs

## === LP Solution Summary ===

## Total materials used: 52 out of 60

## Materials with non-zero allocation: R1, R2, R3, R4, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R1

## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.1.

## === Quality Deviations from Targets ===
```

Table 7: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	0.5000000	0.50000000	0.05429705	0.5000000	0.4800000	0.19873016
B2	0.5000000	0.50000000	0.50000000	0.5000000	0.5000000	0.12126984
B3	0.3736758	0.50000000	0.50000000	0.1152129	0.3736758	0.06666667
B4	0.5000000	0.43809524	0.24592593	0.1150794	0.5000000	0.24000000
B5	0.5000000	0.32182063	0.50000000	0.5000000	0.5000000	0.10000000
B6	0.5000000	0.07333333	0.10666667	0.1400000	0.5000000	0.16000000
B7	0.3977401	0.50000000	0.50000000	0.5000000	0.3977401	0.23728814
B8	0.5000000	0.25000000	0.50000000	0.3000000	0.5000000	0.00000000
B9	0.2333333	0.16666667	0.50000000	0.2333333	0.2333333	0.40000000
B10	0.0000000	0.37500000	0.00000000	0.0000000	0.0000000	0.50000000

4.3 Visualisation





5 Approach 2: Cost Minimisation using Simulated Annealing

Simulated Annealing is a metaheuristic that explores the solution space probabilistically, accepting worse solutions with decreasing probability as the algorithm "cools down."

5.1 Implementation

```
## Running Simulated Annealing...
```

```
## === SA Solution Results ===
```

```
## Best Total Cost: £10871.68
```

```
## Difference from LP: 19.63%
```

```
## === SA Solution Summary ===
```

```
## Total materials used: 60 out of 60
```

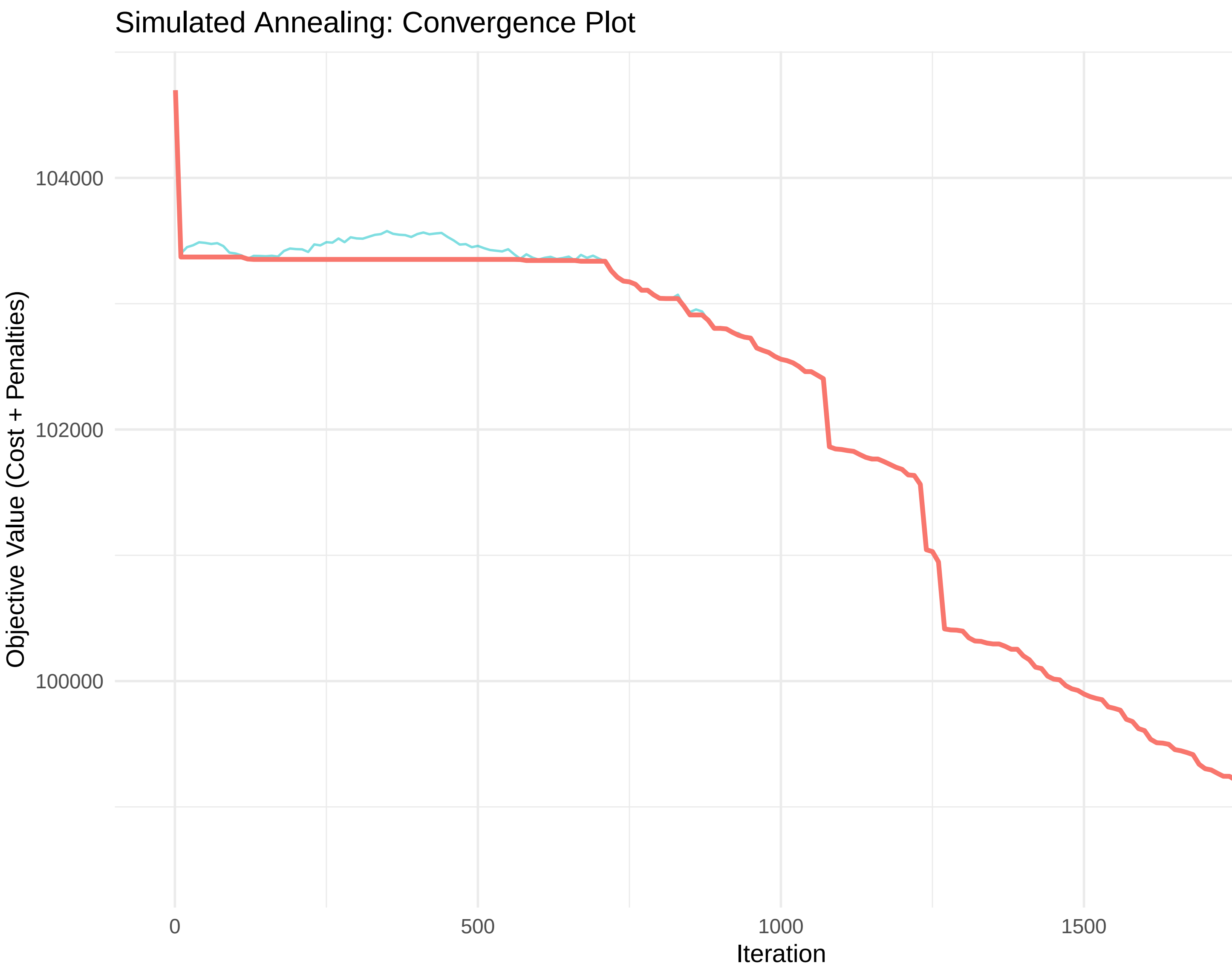
```
## Materials with non-zero allocation: R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15
```

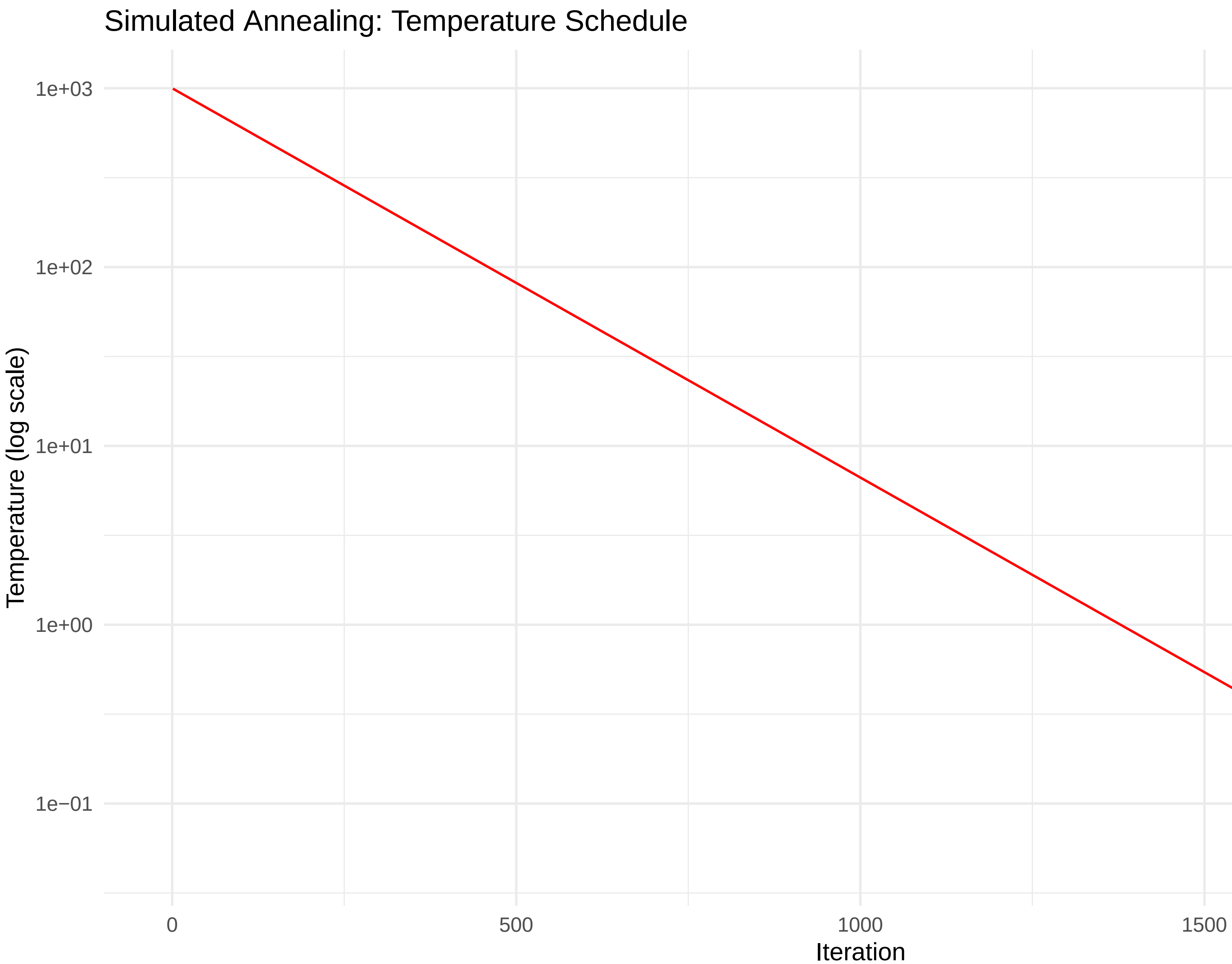
```
## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.2.
## === Quality Deviations from Targets ===
```

Table 8: Absolute deviations from target quality scores

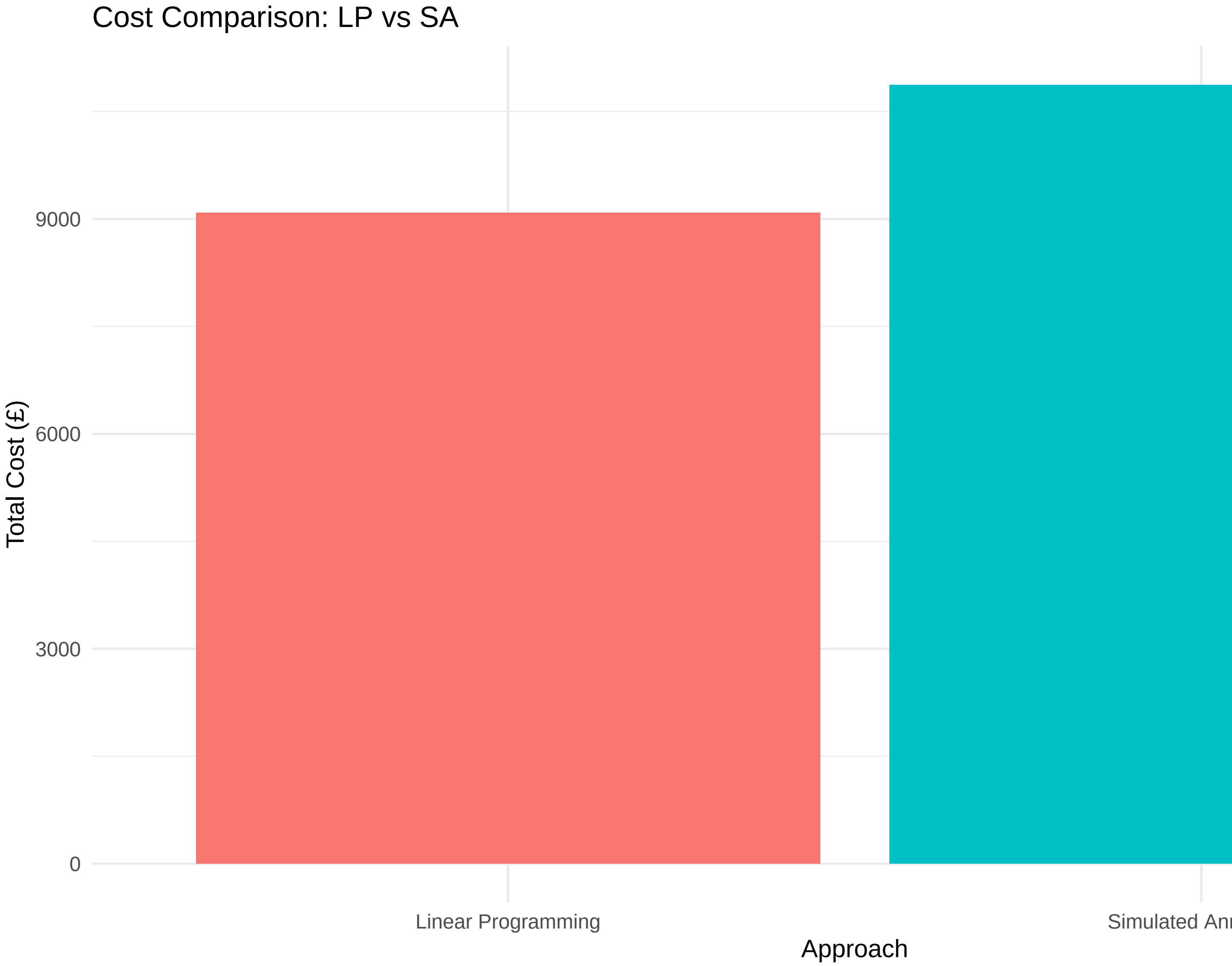
Blend	C1	C2	C3	C4	C5	C6
B1	2.7866403	0.5895314	1.3979100	3.0079163	2.7666403	0.2358893
B2	1.5231095	0.6403009	2.3173849	2.3544282	1.5231095	0.4408001
B3	1.2404905	1.2649454	1.5113435	1.5097660	1.2404905	0.3393983
B4	1.0386832	1.4307854	0.4672402	1.4187900	1.0386832	0.5453917
B5	0.5497852	1.0662379	0.1088752	0.2632299	0.5497852	0.4964167
B6	3.0177087	0.9386653	1.8977966	2.1079642	3.0177087	0.4628573
B7	1.3433559	3.3736074	3.6360283	0.4695635	1.3433559	0.3194533
B8	0.5576406	2.8464602	4.1558419	0.9858363	0.5576406	0.3872241
B9	1.8300274	0.5204099	4.3706506	0.8496498	1.8300274	0.5640213
B10	0.6760079	1.5877678	1.1123896	3.0850628	0.6760079	7.4438965

5.2 Visualisation





Simulated Annealing: Temperature Schedule
1e+03
1e+02
1e+01
1e+00
1e−01
Temperature (log scale)
0
500
1000
1500
Iteration



6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming

Chebyshev Goal Programming minimises the worst-case deviation from multiple goals, providing a balanced solution that trades off cost for quality control.

6.1 Model Formulation

Objective Function: Minimise worst-case deviation

$$\min \max\{w_0 \cdot d_0, w_{jk} \cdot d_{jk}\}$$

Where: - d_0: Deviation from cost goal - d_{jk}: Deviation from quality goal for blend j, characteristic k

Constraints: - All LP constraints - Goal constraints for cost and quality

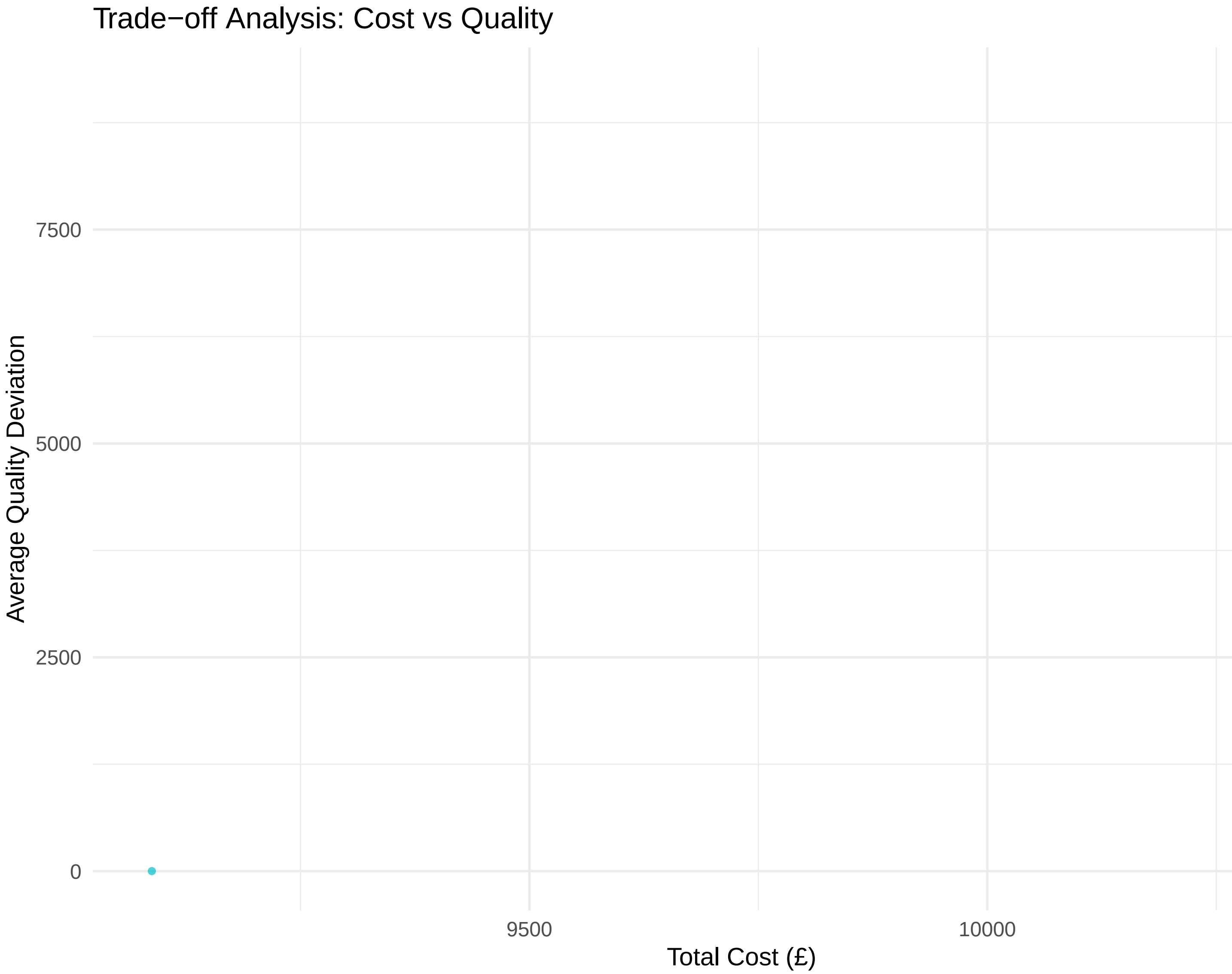
6.2 Implementation

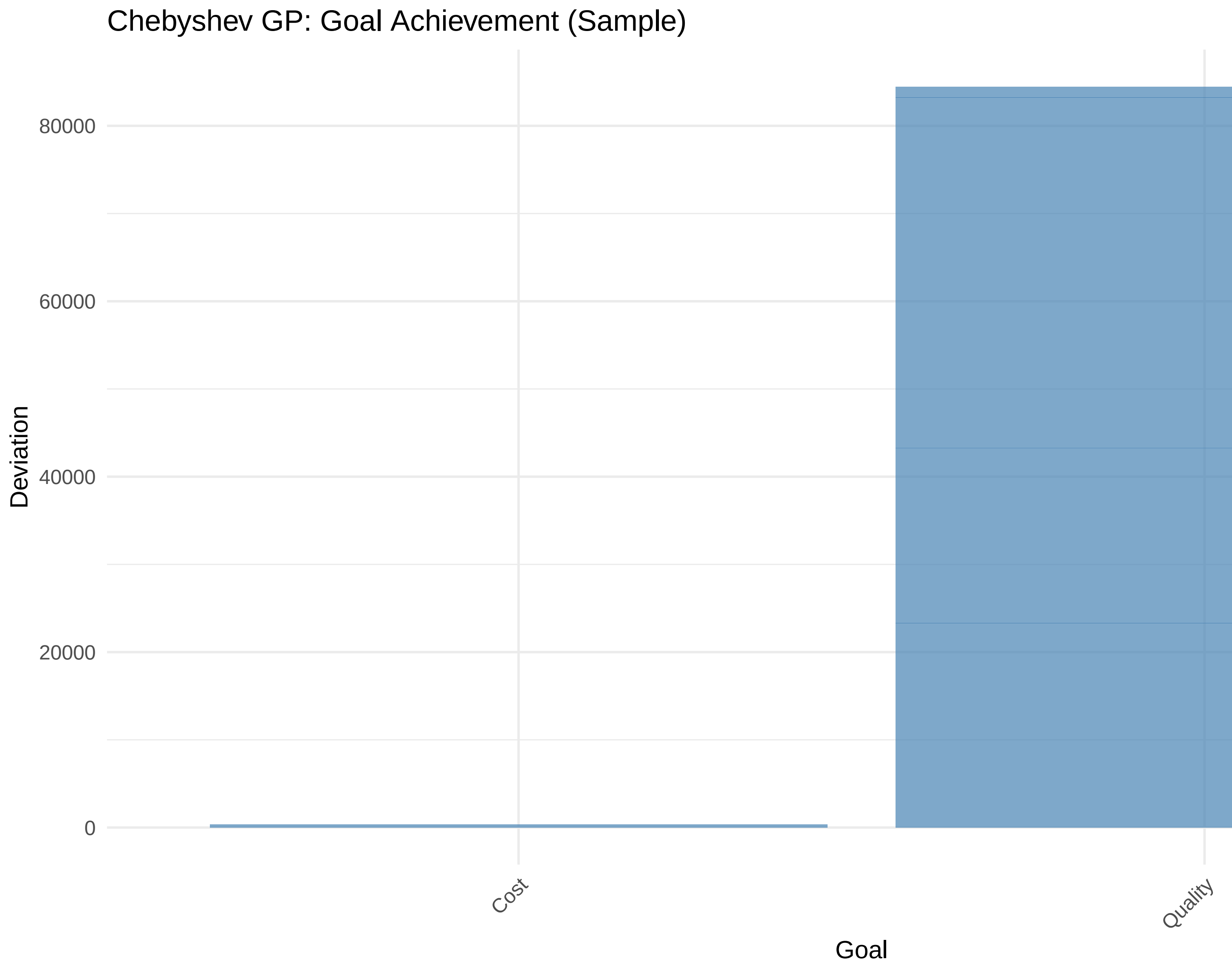
```
## Solving Chebyshev Goal Programming model...

## === Chebyshev GP Solution Results ===

## Worst-case Deviation (z): 39945.0309

## Total Cost: £10375.59

## Cost Deviation from Target: £379.13

## === Chebyshev GP Solution Summary ===

## Total materials used: 48 out of 60

## Materials with non-zero allocation: R2, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R16, R17, R

## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.3.

## === Quality Deviations from Targets ===
```

Table 9: Quality deviations (minimised in worst-case sense)

Blend	C1	C2	C3	C4	C5	C6
B1	3,324.17181172407435952	19,972.515	6,657.505	19,972.515	5,324.17181172407345002	0.00
B2	19,972.51543517221944057	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B3	0.00000000002763036	19,972.515	0.000	16,297.229	0.00000000002763036	0.00
B4	19,972.51543517226309632	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B5	19,972.51543517226673430	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B6	19,972.51543517226309632	1,761.103	16,742.083	19,972.515	19,972.51543517226673430	4,000.00
B7	0.00000000000000000	0.000	0.000	0.000	0.00000000000000000	0.00
B8	0.00000000000000000	0.000	0.000	5,500.000	0.00000000000000000	0.00
B9	1,250.00000000000000000	500.000	1,000.000	0.000	1,250.00000000000000000	0.00
B10	0.00000000000000000	0.000	0.000	821.115	0.00000000000000000	6,568.92

6.3 Visualisation





7 Approach 4: Genetic Algorithms

Genetic Algorithms use evolutionary principles to evolve a population of solutions toward better fitness.

7.1 Implementation

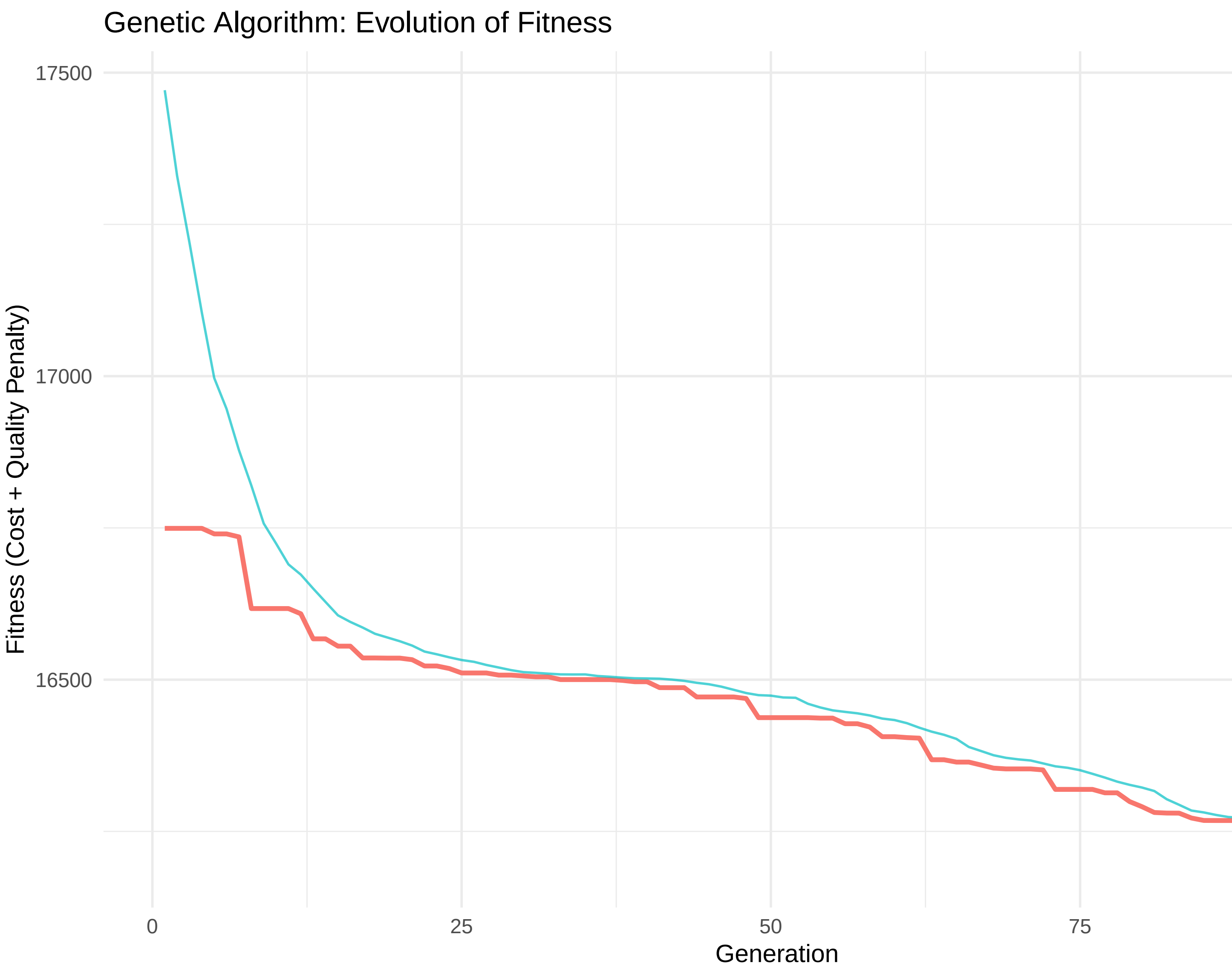
```
## Running Genetic Algorithm...
## === GA Solution Results ===
## Best Total Cost: £15298.75
## Average Quality Deviation: 1.4834
## === GA Solution Summary ===
## Total materials used: 60 out of 60
## Materials with non-zero allocation: R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15
## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.4.
```

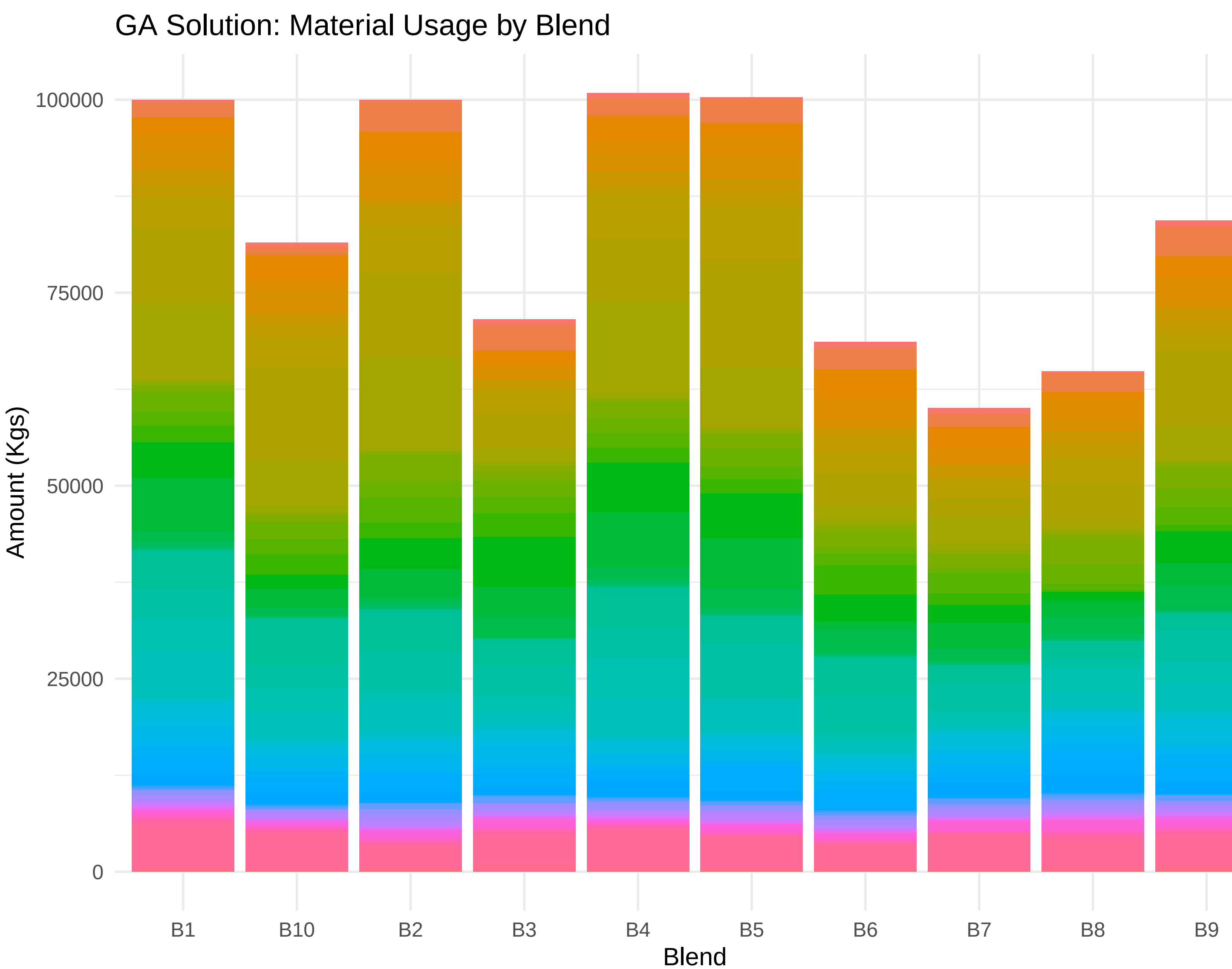
```
## === Quality Deviations from Targets ===
```

Table 10: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.61729937	0.7712157	1.2904388	3.30186139	2.59729937	0.1394708
B2	0.88821608	1.2268571	1.4696659	2.05042342	0.88821608	0.3534963
B3	1.54768944	1.0259450	1.6691668	1.46189268	1.54768944	0.2577706
B4	1.36675696	1.4217326	0.9755323	0.99513973	1.36675696	0.2451125
B5	0.04009265	1.4409208	0.1968494	0.05821596	0.04009265	0.2070019
B6	3.20105056	0.6007810	1.7540294	1.88410720	3.20105056	0.3890987
B7	0.92221389	3.4694649	3.4101175	0.17243708	0.92221389	0.4028456
B8	0.88867299	2.9840725	3.4201390	0.68497989	0.88867299	0.5443162
B9	2.08886197	0.2061741	4.4837665	0.84021615	2.08886197	0.3711146
B10	0.87090839	1.6895563	1.4692887	2.99223185	0.87090839	7.8309674

7.2 Visualisation





8 Summary of Results and Comparative Analysis

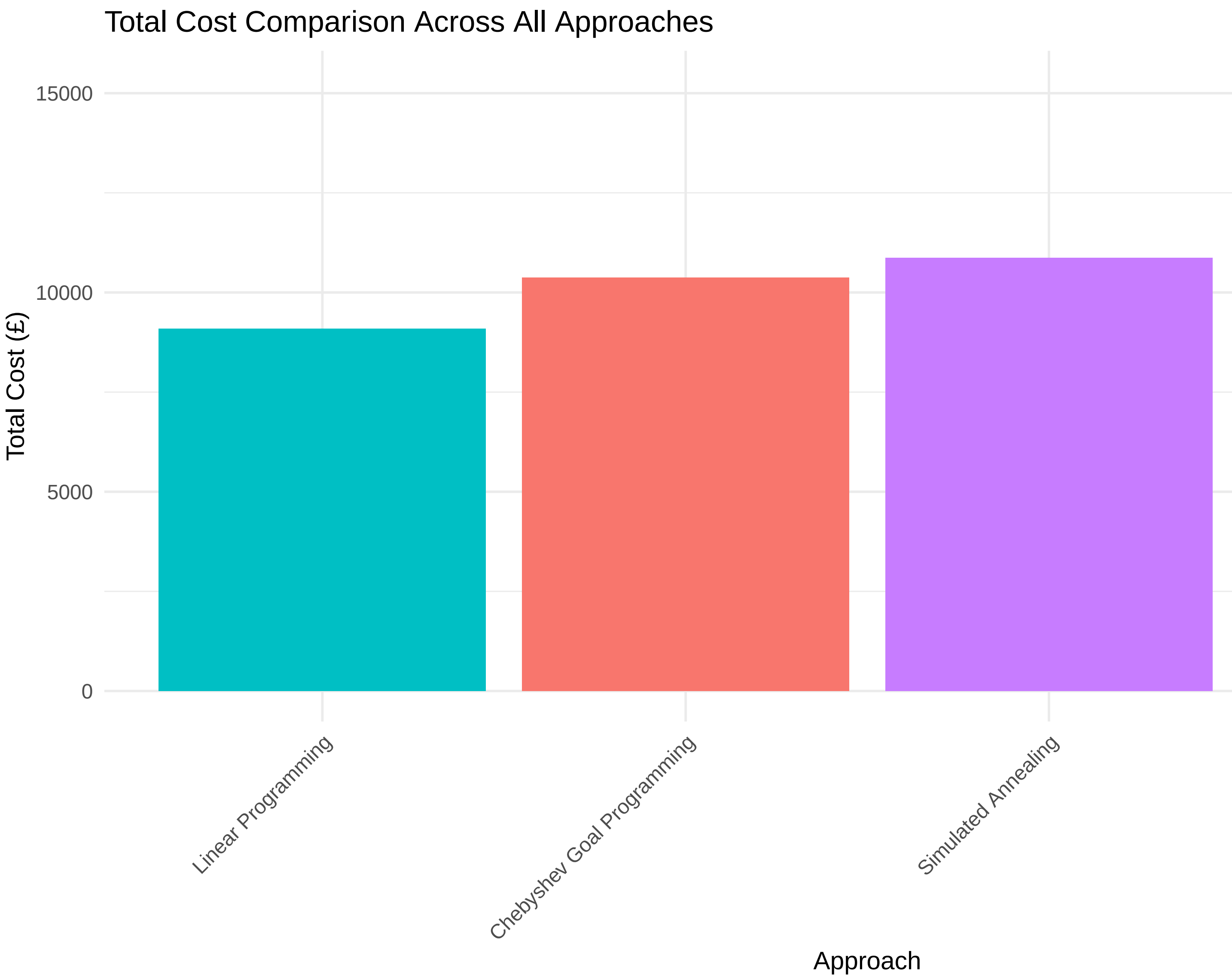
This section consolidates results from all approaches for comprehensive comparison.

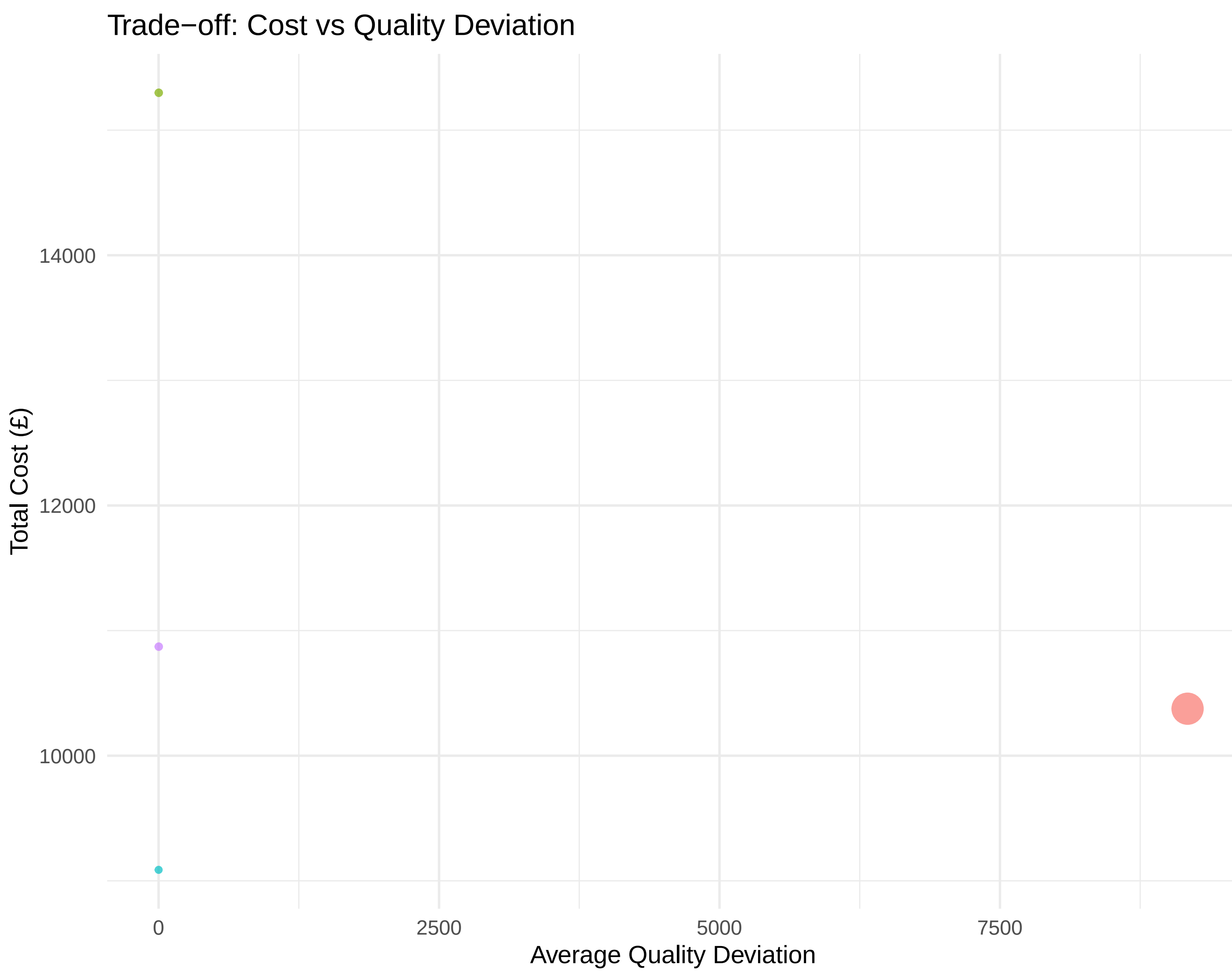
8.1 Comparative Metrics

```
## === Comprehensive Comparison of All Approaches ===
```

Table 11: Performance Metrics Comparison

Approach	Total_Cost	Avg_Quality_Dev	Max_Quality_Dev	Cost_vs_LP_pct
Linear Programming	9,087.691	0.3324814	0.500000	0.00000
Simulated Annealing	10,871.683	1.5271438	7.443897	19.63086
Chebyshev Goal Programming	10,375.592	9,172.2778281	19,972.515435	14.17193
Genetic Algorithm	15,298.745	1.4833656	7.830967	68.34580


Total Cost Comparison Across All Approaches
15000
10000
5000
0
Total Cost (£)
Linear Programming
Chebyshev Goal Programming
Simulated Annealing
Approach



8.2 Key Findings

```
## === Key Findings ===
## 1. Cost Performance:
##    - LP provides the optimal cost baseline: £9087.69
##    - SA achieves 119.63% of LP cost
##    - Chebyshev GP trades 14.17% cost increase for better quality control
##    - GA achieves 168.35% of LP cost
## 2. Quality Performance:
##    - Best average quality: Linear Programming (deviation: 0.3325)
## 3. Recommendation:
##    - Linear Programming provides the best cost-performance trade-off.
##    - For near-optimal solutions quickly, Simulated Annealing is valuable.
```

8.3 Solution Matrices Comparison

```
## === Solution Matrix Comparison (Sample: Blend 1) ===
```

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA	Chebyshev_GP	GA
R1	0.000	685.75414	0.000	345.26263
R2	0.000	328.14362	0.000	46.01968
R3	0.000	274.23998	0.000	1,883.96722
R4	0.000	209.27734	0.000	28.72304
R5	0.000	5,824.71044	0.000	2,041.55278
R6	0.000	1,653.47115	0.000	2,097.19686
R7	0.000	5,075.37961	0.000	2,633.32234
R8	0.000	154.74357	0.000	101.26902
R9	0.000	77.86111	0.000	1,971.07498
R10	0.000	2,577.00491	0.000	1,524.54605
R11	40,000.000	6,079.99901	40,000.000	4,102.36677
R12	7,554.827	4,178.83026	32,528.357	9,580.97801
R13	0.000	2,708.85252	0.000	10,022.46470
R14	5,000.000	1,165.54908	0.000	251.62075
R15	0.000	34.72603	0.000	413.83827
R16	0.000	1,505.94895	0.000	776.06527
R17	0.000	5,405.12814	0.000	2,612.12687
R18	0.000	3,398.58917	3,328.753	1,771.55404
R19	0.000	78.54172	0.000	2,202.44543
R20	4,371.720	5,155.58591	0.000	4,560.85982
R21	0.000	3,310.24018	0.000	6,999.97026
R22	0.000	430.88646	0.000	1,335.34805
R23	0.000	448.81430	0.000	878.01965
R24	0.000	245.08577	0.000	46.87489
R25	0.000	108.50489	0.000	66.37919
R26	0.000	221.72502	0.000	107.13074
R27	0.000	4,136.75719	0.000	4,957.73805
R28	0.000	4,268.49254	0.000	3,819.58242
R29	0.000	4,130.67317	0.000	4,465.99533
R30	0.000	3,043.38092	0.000	5,786.31102
R31	0.000	1,605.28772	0.000	368.77536

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA	Chebyshev_GP	GA
R32	0.000	87.44411	0.000	639.41652
R33	0.000	3,777.89431	0.000	1,641.49304
R34	0.000	2,826.32385	0.000	1,152.46295
R35	8,769.679	1,615.73488	0.000	1,629.91522
R36	0.000	1,912.36497	0.000	1,155.59974
R37	0.000	3,652.95923	0.000	1,376.22025
R38	20,000.000	4,529.42092	14,133.729	2,194.37472
R39	0.000	216.52602	0.000	1,300.47274
R40	0.000	97.51660	0.000	327.40451
R41	0.000	1,455.47640	0.000	181.76639
R42	1,000.000	197.29215	0.000	99.12533
R43	0.000	290.09395	0.000	681.95288
R44	4,968.254	32.87680	0.000	539.92188
R45	0.000	502.32180	0.000	385.55380
R46	0.000	352.67755	0.000	287.61705
R47	2,000.000	205.87755	0.000	191.51662
R48	0.000	38.52968	0.000	109.31091
R49	0.000	354.97703	0.000	140.15371
R50	0.000	114.10703	0.000	107.30440
R51	0.000	320.06324	0.000	15.57253
R52	0.000	23.46473	0.000	247.28402
R53	0.000	1,067.98030	0.000	694.62497
R54	0.000	51.63754	0.000	39.42455
R55	0.000	201.38365	0.000	77.95744
R56	0.000	102.76081	0.000	136.74957
R57	0.000	1,457.63147	0.000	747.67437
R58	6,335.520	5,105.47192	10,009.162	5,531.67326
R59	0.000	1,238.10303	0.000	443.93226
R60	0.000	262.03791	0.000	124.14491

9 Conclusion

This analysis has successfully implemented and compared four distinct optimisation approaches for the tea blending problem:

1. **Linear Programming** provides the mathematically optimal cost solution, serving as an essential

benchmark. However, it treats quality deviations as hard constraints within tolerance, not as objectives to minimise.

2. **Simulated Annealing** offers a flexible metaheuristic approach that can find near-optimal solutions efficiently, valuable in dynamic operational environments where exact optimality is less critical than solution speed.

3. **Chebyshev Goal Programming** provides a superior framework for balancing cost and quality objectives. By formally incorporating decision-maker preferences through goals and weights, it delivers balanced, actionable solutions that directly manage the trade-off between production cost and product quality.

4. **Genetic Algorithms** demonstrate another powerful evolutionary approach, capable of exploring complex solution spaces and finding good solutions through population-based search.

The comparative analysis reveals that while LP provides the lowest cost, multi-objective approaches like Chebyshev Goal Programming are better aligned with business realities where maintaining brand integrity through consistent quality characteristics is as important as minimising cost.

10 Appendix: Code Execution Notes

10.1 Required Libraries

All required R packages can be installed using:

```r
install.packages(c("dplyr", "Rglpk", "ompr", "ompr.roi", "ROI", "ROI.plugin.glpk",
                   "lpSolveAPI", "ggplot2", "GA", "knitr", "flextable", "tidyr"))
```

10.2 Reproducibility

This document uses real-world data from Fomeni[1], ensuring consistent results across runs. Stochastic algorithms (Simulated Annealing and Genetic Algorithms) may produce slightly different results on each run due to their probabilistic nature, but the underlying data and deterministic algorithms (Linear Programming and Goal Programming) will produce identical results.

10.3 Parameter Tuning

Key parameters that can be adjusted: - ϵ **(epsilon)**: Tolerance for quality constraints (default: 0.5 for Fomeni data) - **SA parameters**: Initial temperature, cooling rate, iterations - **GA parameters**: Population size, generations, mutation/crossover rates - **Goal Programming weights**: Cost vs quality priorities

11 Appendix: Complete Recipe Matrices

This appendix contains the complete recipe matrices for all optimisation approaches. Each matrix shows the amount (in kg) of each raw material used in each blend.

11.1 Linear Programming Solution

```
## === LP Solution: Complete Recipe Matrix (x_ij in Kgs) ===
```

Table 13: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R2	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R3	0.000	0.00000	20,450.410	0.000	4,549.590	0.0000	0.0000	0.0000	0.000	0
R4	0.000	0.00000	0.000	0.000	1,000.000	0.0000	0.0000	0.0000	0.000	0
R5	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R6	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	6,333.333	0
R7	0.000	0.00000	0.000	20,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R8	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R9	0.000	0.00000	2,207.300	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R10	0.000	0.00000	0.000	0.000	0.000	15,000.0000	0.0000	0.0000	0.000	0
R11	40,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R12	7,554.827	50,277.77778	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R13	0.000	0.00000	0.000	4,164.021	0.000	0.0000	0.0000	0.0000	0.000	0
R14	5,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R15	0.000	0.00000	0.000	0.000	0.000	0.0000	2,333.3333	2,666.6667	0.000	0
R16	0.000	0.00000	0.000	0.000	1,250.000	0.0000	0.0000	0.0000	0.000	18,750
R17	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	833.3333	0.000	0
R18	0.000	20,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R19	0.000	3,746.03175	0.000	16,253.968	0.000	0.0000	0.0000	0.0000	0.000	0
R20	4,371.720	0.00000	11,543.614	0.000	24,084.666	0.0000	0.0000	0.0000	0.000	0
R21	0.000	0.00000	0.000	0.000	13,384.326	0.0000	0.0000	0.0000	0.000	0
R22	0.000	0.00000	0.000	19,666.667	0.000	333.3333	0.0000	0.0000	0.000	0
R23	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R24	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R25	0.000	0.00000	0.000	500.000	0.000	0.0000	0.0000	0.0000	0.000	0
R26	0.000	0.00000	1,000.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R27	0.000	0.00000	19,170.041	6,082.011	14,747.948	0.0000	0.0000	0.0000	0.000	0
R28	0.000	0.00000	0.000	7,333.333	0.000	32,666.6667	0.0000	0.0000	0.000	0
R29	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R30	0.000	0.00000	0.000	0.000	0.000	0.0000	338.9831	0.0000	0.000	0
R31	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R32	0.000	0.00000	3,333.333	0.000	0.000	0.0000	0.0000	0.0000	1,666.667	0
R33	0.000	0.00000	0.000	0.000	0.000	0.0000	4,615.8192	0.0000	0.000	0

Table 13: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R34	0.000	0.00000	0.000	0.000	9,500.000	0.0000	0.0000	500.0000	0.000	0
R35	8,769.679	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R36	0.000	7,523.80952	0.000	0.000	0.000	0.0000	2,118.6441	0.0000	0.000	0
R37	0.000	0.00000	0.000	10,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R38	20,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R39	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R40	0.000	0.00000	0.000	0.000	2,000.000	0.0000	0.0000	0.0000	0.000	0
R41	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R42	1,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R43	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R44	4,968.254	31.74603	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R45	0.000	0.00000	0.000	0.000	1,489.625	0.0000	0.0000	0.0000	0.000	0
R46	0.000	0.00000	0.000	0.000	0.000	2,000.0000	0.0000	0.0000	0.000	0
R47	2,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R48	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R49	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R50	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R51	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R52	0.000	0.00000	0.000	0.000	5,000.000	0.0000	0.0000	0.0000	0.000	0
R53	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R54	0.000	0.00000	0.000	0.000	0.000	0.0000	593.2203	0.0000	0.000	0
R55	0.000	0.00000	0.000	1,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R56	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	1,000.0000	0.000	0
R57	0.000	0.00000	2,295.301	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R58	6,335.520	9,420.63492	0.000	0.000	22,993.845	0.0000	0.0000	0.0000	0.000	1,250
R59	0.000	5,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R60	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0

11.2 Simulated Annealing Solution

```
## === SA Solution: Complete Recipe Matrix (x_ij in Kgs) ===
```

Table 14: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R1	685.75414	1,833.61992	224.216310	703.392332	642.735987	218.696697	50.613900	87.7587172	140.01829
R2	328.14362	79.80091	9.663633	193.314765	53.995047	145.769303	23.813552	16.6469091	41.53614
R3	274.23998	3,270.34722	123.591095	5,078.524167	5,099.797460	1,454.442965	357.835281	317.1694345	129.83233
R4	209.27734	211.56485	126.238028	194.807005	88.692509	38.784616	6.366333	11.2261075	13.54066
R5	5,824.71044	5,273.18086	1,091.602834	5,868.005028	1,563.896861	2,512.360856	356.938629	363.4109218	602.82244
R6	1,653.47115	5,267.29640	2,822.300214	4,690.262645	1,131.090890	1,920.253653	128.984281	411.2740906	685.31407
R7	5,075.37961	3,369.05574	848.672632	1,588.735104	5,469.147431	2,161.669178	265.729166	147.1196010	400.75147
R8	154.74357	223.81968	119.075307	103.039129	229.710636	70.038855	16.947892	15.0416668	7.58957
R9	77.86111	608.80160	3,377.986716	3,633.432224	1,209.767295	2,923.003350	130.134552	81.7385596	24.25519
R10	2,577.00491	4,014.26839	1,936.964900	2,341.203806	3,078.253589	311.705706	207.335266	217.3556941	86.86220
R11	6,079.99901	3,624.55578	2,869.420758	3,120.269766	4,198.840358	900.530256	193.166318	186.1812155	319.43835
R12	4,178.83026	1,729.75971	985.885476	3,306.137441	101.640122	930.753189	534.108089	111.1369038	609.63873
R13	2,708.85252	1,320.78383	1,804.958163	1,949.708507	2,450.685396	773.287177	209.726702	136.9602800	674.35576
R14	1,165.54908	1,764.21262	553.771771	815.372507	96.719114	173.924530	28.538475	0.3064384	9.38587
R15	34.72603	591.73254	860.644565	1,526.588110	1,329.110040	26.184318	166.118354	55.6511133	62.40432
R16	1,505.94895	3,007.56944	1,568.317409	4,810.269858	4,902.922643	2,018.802713	217.456260	119.5785789	485.60930
R17	5,405.12814	165.93536	2,236.439244	4,956.076099	5,266.069388	429.485617	369.621080	129.3090818	533.83338
R18	3,398.58917	4,596.63764	3,041.220295	267.719980	6,626.928883	1,119.444526	60.236782	71.2911327	397.15585
R19	78.54172	4,738.06090	3,537.969655	4,993.947878	98.829122	809.690174	689.667012	31.8067227	56.03598
R20	5,155.58591	1,400.45071	821.565921	3,806.619177	535.479908	3,956.516443	369.469121	169.3572628	82.78316
R21	3,310.24018	4,866.75543	2,458.367637	4,006.816186	2,773.225638	2,749.671252	526.693089	99.4486444	247.97979
R22	430.88646	1,290.48518	2,455.808050	2,193.375295	7,792.019959	1,696.401505	691.551260	23.8898797	306.79890
R23	448.81430	733.00318	449.585173	628.741316	1,879.587998	418.131491	36.354254	86.4086802	78.37426
R24	245.08577	200.43916	156.685654	34.588260	120.060777	177.666723	12.597080	1.7296471	27.95814
R25	108.50489	55.09395	29.093543	105.267087	75.809728	58.976373	11.220401	3.5076684	8.70262
R26	221.72502	156.91686	83.519092	139.705514	285.592160	10.972616	17.954578	2.6554728	11.95035
R27	4,136.75719	879.89769	2,284.478342	3,968.274264	558.902139	1,739.733402	408.526626	13.9488924	626.20961
R28	4,268.49254	6,641.57031	2,187.499038	4,226.306055	3,234.284104	355.234697	91.091978	4.3644513	155.71036
R29	4,130.67317	5,805.97717	1,584.698825	1,858.086302	4,387.445015	1,469.562679	657.445096	164.6603167	93.38182
R30	3,043.38092	3,426.60608	2,738.312761	4.620828	4,079.489441	1,363.529864	304.880266	205.9131055	698.46549
R31	1,605.28772	611.21213	289.698080	1,152.151505	932.332031	29.697713	154.660679	30.8860559	21.08495
R32	87.44411	1,190.00742	355.252314	247.022200	1,817.995796	709.762591	144.864869	32.0376013	104.67353
R33	3,777.89431	590.52756	1,328.844297	459.744381	2,321.660149	838.923665	5.641323	103.3120178	58.21080

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R34	2,826.32385	1,352.77448	1,504.517466	896.559644	1,636.125890	984.455180	169.565414	215.9012530	134.54739
R35	1,615.73488	1,731.96154	236.436206	1,417.766931	1,115.377421	1,936.594382	268.461988	98.6363260	139.11717
R36	1,912.36497	1,353.31221	2,260.771997	756.260471	573.103541	2,162.669968	302.769049	156.4090760	294.40502
R37	3,652.95923	1,648.92975	50.522077	908.726062	217.542052	2,545.198201	263.784415	96.3543095	230.98940
R38	4,529.42092	4,834.54923	832.025396	4,934.885710	3,320.983618	135.054774	48.209182	59.4809231	297.90620
R39	216.52602	4,924.82263	1,916.484301	1,639.028781	2,927.633238	2,024.198062	431.013865	147.7071445	158.13288
R40	97.51660	457.06632	271.544260	21.177942	605.553321	244.055235	78.782270	34.4476106	35.21161
R41	1,455.47640	856.05457	184.255882	106.268219	1,647.945534	324.525124	1.831330	68.9068638	78.05674
R42	197.29215	279.75077	110.592794	299.757667	13.929188	1.000957	28.727069	15.3020249	15.82153
R43	290.09395	707.79932	579.189676	1,325.672479	1,085.163936	736.429670	144.904021	31.6526348	34.60844
R44	32.87680	934.14768	454.566113	1,522.469913	876.595426	683.042169	75.131200	30.7644159	142.73560
R45	502.32180	1,013.65727	789.292870	1,203.947663	402.248629	359.637071	117.135585	104.0540066	165.80753
R46	352.67755	558.07711	347.480889	154.512208	12.959484	229.562888	63.549706	40.5384615	39.51681
R47	205.87755	225.17460	208.879880	614.370771	350.293691	67.393105	78.751884	33.7887621	6.58674
R48	38.52968	306.85984	113.877957	283.045999	90.965212	67.359300	23.858871	8.7612622	3.79216
R49	354.97703	70.90887	26.819178	342.081461	5.160512	16.135274	26.490659	26.2853457	26.51510
R50	114.10703	197.98246	149.129467	274.861558	30.276243	154.380518	7.892182	7.2301325	13.24886
R51	320.06324	141.52729	177.120249	176.744200	30.379875	68.389590	31.070365	12.1255596	9.48146
R52	23.46473	82.81335	1,205.980017	1,831.187568	244.028242	1,135.778823	18.750736	164.9649422	144.08732
R53	1,067.98030	498.09201	647.026658	702.651310	1,126.747238	428.897916	57.653379	40.0021152	33.48704
R54	51.63754	361.98088	75.521342	201.752181	61.196632	102.442253	25.983998	10.5572246	0.47527
R55	201.38365	209.18700	93.161150	210.723154	164.587524	60.860717	3.099907	8.2575442	20.49582
R56	102.76081	237.40020	82.937727	221.572662	266.952513	24.872415	13.665775	1.7764015	6.12313
R57	1,457.63147	312.20322	192.101083	1,051.021216	760.437317	727.071483	162.498141	8.6734346	112.75681
R58	5,105.47192	2,849.20108	1,759.140311	5,224.716019	7,398.027484	453.535689	3.214343	43.0914750	90.98879
R59	1,238.10303	720.84905	493.853854	685.846297	635.849577	767.747134	95.183998	82.6772245	104.50591
R60	262.03791	58.70219	101.570176	161.446779	101.262912	173.516865	16.829558	0.5934578	32.98962

11.3 Chebyshev Goal Programming Solution

```
## === Chebyshev GP Solution: Complete Recipe Matrix (x_ij in Kgs) ===
```

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R2	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R3	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R4	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R5	0.000	0.0000	1,234.6787	0.0000	0.0000	0.000	0	0	0	0.0000
R6	0.000	0.0000	7,958.0429	0.0000	12,041.9571	0.000	0	0	0	0.0000
R7	0.000	0.0000	8,359.9383	0.0000	0.0000	1,890.062	0	0	9,750	0.0000
R8	0.000	0.0000	0.0000	0.0000	0.0000	1,000.000	0	0	0	0.0000
R9	0.000	102.3406	0.0000	14,897.6594	0.0000	0.000	0	0	0	0.0000
R10	0.000	0.0000	0.0000	15,000.0000	0.0000	0.000	0	0	0	0.0000
R11	40,000.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R12	32,528.357	38,998.6023	0.0000	0.0000	8,473.0411	0.000	0	0	0	0.0000
R13	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	1,100	0	0.0000
R14	0.000	0.0000	0.0000	0.0000	0.0000	0.000	5,000	0	0	0.0000
R15	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R16	0.000	1,231.6725	0.0000	0.0000	0.0000	0.000	0	0	0	18,768.3275
R17	0.000	0.0000	0.0000	0.0000	0.0000	3,082.454	3,000	3,900	0	0.0000
R18	3,328.753	0.0000	0.0000	0.0000	16,671.2474	0.000	0	0	0	0.0000
R19	0.000	0.0000	0.0000	0.0000	20,000.0000	0.000	0	0	0	0.0000
R20	0.000	0.0000	0.0000	40,000.0000	0.0000	0.000	0	0	0	0.0000
R21	0.000	18,417.3750	18,134.9499	3,447.6751	0.0000	0.000	0	0	0	0.0000
R22	0.000	0.0000	0.0000	7,615.9712	7,509.4432	0.000	0	0	0	0.0000
R23	0.000	0.0000	0.0000	5,000.0000	0.0000	0.000	0	0	0	0.0000
R24	0.000	0.0000	0.0000	0.0000	516.0415	0.000	0	0	0	0.0000
R25	0.000	0.0000	0.0000	0.0000	500.0000	0.000	0	0	0	0.0000
R26	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R27	0.000	0.0000	21,460.7884	0.0000	0.0000	14,027.485	0	0	0	0.0000
R28	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R29	0.000	0.0000	0.0000	0.0000	0.0000	25,000.000	0	0	0	0.0000
R30	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R31	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R32	0.000	0.0000	0.0000	0.0000	0.0000	5,000.000	0	0	0	0.0000
R33	0.000	0.0000	0.0000	0.0000	0.0000	0.000	2,000	0	0	0.0000

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R34	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R35	0.000	0.0000	0.0000	0.0000	624.2942	0.000	0	0	0	0.0000
R36	0.000	4,571.0624	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R37	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R38	14,133.729	0.0000	0.0000	1,902.9377	0.0000	0.000	0	0	0	0.0000
R39	0.000	0.0000	0.0000	0.0000	15,000.0000	0.000	0	0	0	0.0000
R40	0.000	2,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R41	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R42	0.000	0.0000	339.4425	0.0000	0.0000	0.000	0	0	250	410.5575
R43	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R44	0.000	0.0000	0.0000	522.9126	4,477.0874	0.000	0	0	0	0.0000
R45	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R46	0.000	0.0000	0.0000	2,000.0000	0.0000	0.000	0	0	0	0.0000
R47	0.000	0.0000	0.0000	0.0000	2,000.0000	0.000	0	0	0	0.0000
R48	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R49	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R50	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R51	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	821.1150
R52	0.000	2,487.8407	2,512.1593	0.0000	0.0000	0.000	0	0	0	0.0000
R53	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R54	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R55	0.000	529.7838	0.0000	470.2162	0.0000	0.000	0	0	0	0.0000
R56	0.000	754.9784	0.0000	245.0216	0.0000	0.000	0	0	0	0.0000
R57	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R58	10,009.162	28,906.3443	0.0000	0.0000	1,084.4942	0.000	0	0	0	0.0000
R59	0.000	0.0000	0.0000	3,897.6061	1,102.3939	0.000	0	0	0	0.0000
R60	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

11.4 Genetic Algorithm Solution

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## === GA Solution: Complete Recipe Matrix (x_ij in Kgs) ===
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Table 16: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R1	345.26263	255.70991	657.69329	644.14176	268.25613	764.57114	668.97459	245.17530	701.7545
R2	46.01968	63.88141	116.25685	83.69711	66.88601	134.74529	141.26456	37.11606	142.4083
R3	1,883.96722	3,794.92973	3,270.40682	2,099.85860	2,919.37039	2,592.44459	1,669.28087	2,309.06271	3,615.3142
R4	28.72304	107.08479	31.21420	193.77946	92.21013	123.22342	20.67909	113.56729	165.8387
R5	2,041.55278	3,579.60478	1,541.39932	3,189.45875	1,566.97339	3,596.47291	2,714.45396	1,088.95549	2,621.9722
R6	2,097.19686	1,957.58274	815.07581	1,542.59504	2,600.81457	2,593.87465	1,813.79511	2,341.19934	3,065.2130
R7	2,633.32234	3,226.57810	1,310.51665	2,429.45814	3,020.48930	1,295.14377	383.09259	1,584.02570	1,101.1018
R8	101.26902	85.58577	29.59746	134.72404	153.41997	42.13146	159.59656	83.93262	61.6029
R9	1,971.07498	950.20578	1,068.17843	1,895.61837	1,821.16462	1,211.52988	1,350.04430	1,625.45577	1,742.5106
R10	1,524.54605	2,399.57310	573.39793	848.93568	1,851.11165	2,109.13883	588.14470	1,715.32970	1,650.3178
R11	4,102.36677	6,059.77091	2,974.33234	5,778.55795	6,910.65173	2,617.91033	2,359.50979	3,369.16762	2,077.2277
R12	9,580.97801	10,926.28301	4,314.75502	8,229.91487	13,833.23144	4,216.12912	2,241.95793	5,080.70685	9,691.3586
R13	10,022.46470	11,980.87064	1,814.52419	11,779.04776	7,839.16958	1,826.72446	3,476.88881	1,086.22886	4,485.2953
R14	251.62075	247.21670	426.82208	772.49354	212.23624	630.34650	1,015.52335	296.31947	247.9845
R15	413.83827	325.82109	766.92387	394.93629	528.83987	735.06229	382.73412	712.45935	438.1217
R16	776.06527	3,328.47204	1,174.08555	2,023.54409	1,878.03788	2,114.27345	1,721.55044	3,311.99062	2,877.8445
R17	2,612.12687	2,244.55551	2,107.68788	1,983.97076	2,266.90371	878.97022	649.06149	2,549.22941	2,444.1192
R18	1,771.55404	3,243.00207	2,127.69988	1,973.84362	1,630.36615	1,505.61000	2,724.12323	801.81831	2,310.9262
R19	2,202.44543	2,068.78374	3,098.94963	1,882.99150	1,869.68112	3,743.08674	1,493.29635	252.59023	825.2853
R20	4,560.85982	3,934.15283	6,417.02914	6,446.71597	5,792.00833	3,532.12472	2,262.62299	1,116.27324	4,136.9996
R21	6,999.97026	3,725.09308	3,973.49753	7,112.94129	6,500.58158	1,069.97437	3,336.55354	2,106.42614	2,854.5419
R22	1,335.34805	773.51184	2,669.31649	1,564.23633	2,571.77676	2,967.43127	1,627.25296	2,280.54384	3,241.6262
R23	878.01965	613.30501	31.65478	746.70476	846.45956	322.57369	359.92447	699.39793	217.2875
R24	46.87489	112.74549	77.52102	175.23656	57.43071	128.05064	113.94831	126.26785	125.0557
R25	66.37919	46.26021	57.40878	28.74339	25.90011	74.11754	22.80752	39.69069	74.8653
R26	107.13074	150.43015	134.24255	58.49993	59.89737	56.66516	139.45503	67.26967	63.7800
R27	4,957.73805	5,309.46629	3,314.48065	5,403.51619	3,546.49182	4,925.02556	2,439.54019	2,158.03780	2,133.9143
R28	3,819.58242	5,109.54751	3,832.60100	3,745.52561	6,955.44674	4,930.20419	3,327.25190	1,164.42804	4,131.1522
R29	4,465.99533	1,127.28623	1,956.48817	5,076.93119	551.36511	824.67365	1,435.47659	3,386.24413	2,894.5146
R30	5,786.31102	3,781.54276	1,708.43293	5,040.34342	3,838.87520	1,204.18365	901.18833	1,751.27899	3,157.2494
R31	368.77536	702.39820	545.14698	321.73391	438.67905	690.91733	276.35964	474.29085	617.5133
R32	639.41652	690.42569	635.31467	305.01699	220.67563	350.55319	580.34482	268.97503	609.6386
R33	1,641.49304	316.19439	1,017.75531	1,250.11752	959.98589	507.16472	1,400.48250	837.14799	1,884.9492

Table 16: GA Solution: Amount of Each Material in Each Blend
(60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R34	1,152.46295	1,435.94737	767.96266	345.59992	666.38732	1,242.80093	696.35085	1,146.33838	1,371.8694
R35	1,629.91522	926.43050	1,160.16263	1,399.89020	1,080.11738	386.94385	515.82770	1,183.33038	430.2861
R36	1,155.59974	1,278.58388	1,428.96120	504.67095	717.08917	1,099.01180	1,401.40511	922.07760	982.9226
R37	1,376.22025	394.41586	1,193.22062	589.73985	618.43718	1,111.58016	1,076.09785	1,158.81421	1,482.5143
R38	2,194.37472	2,475.57941	1,538.42590	1,044.01803	3,013.39908	2,337.78196	948.08296	2,828.90672	2,046.9564
R39	1,300.47274	1,302.60950	987.87312	2,186.35852	1,337.29671	185.78415	2,073.51787	2,310.16138	1,671.2788
R40	327.40451	99.71337	141.72582	167.61981	142.27736	279.93840	172.73283	324.53181	92.3510
R41	181.76639	692.73423	838.36785	331.56790	459.68191	438.08448	646.64989	454.91688	671.1867
R42	99.12533	110.84251	60.54627	178.65486	127.06055	35.27367	94.47913	157.53144	17.4791
R43	681.95288	685.74922	235.51364	765.36535	564.91650	521.75637	454.21374	578.67731	414.8660
R44	539.92188	731.72918	635.11424	207.12970	181.15406	699.15073	677.22073	666.57763	375.1467
R45	385.55380	620.25885	521.93496	410.71043	836.58000	317.88130	466.47248	377.96714	561.1841
R46	287.61705	198.76438	233.22048	263.44297	194.76124	124.27709	94.55790	282.29654	237.1813
R47	191.51662	198.70212	34.77397	226.82983	293.49577	123.33421	48.97986	265.30312	337.4628
R48	109.31091	115.27195	118.94459	158.24337	36.97624	136.05796	78.03202	95.20699	80.2633
R49	140.15371	63.89379	38.54256	114.69170	89.16047	143.53642	119.44709	71.04590	115.6657
R50	107.30440	43.31997	136.92818	132.16348	56.11937	141.39709	31.51060	111.20818	169.2831
R51	15.57253	91.33603	62.06923	128.02109	33.33320	43.31025	194.04315	129.32830	186.5876
R52	247.28402	799.22919	638.56119	169.35536	598.53587	539.59891	304.32606	634.21559	701.6121
R53	694.62497	311.25754	574.34449	267.93898	486.47233	287.60090	871.32325	757.87733	498.6152
R54	39.42455	123.70696	139.85904	56.44182	71.52251	151.88615	145.35029	196.23712	33.3924
R55	77.95744	127.57042	116.38093	125.44867	106.28972	56.62198	134.12332	99.50596	26.5234
R56	136.74957	143.13152	57.53620	113.96123	146.95883	26.37424	106.12031	105.91254	74.8796
R57	747.67437	441.26030	517.89374	431.36972	389.68484	320.97694	132.41572	674.71215	771.1575
R58	5,531.67326	2,734.18870	3,895.82821	4,710.27342	3,945.73742	2,917.63619	4,426.36633	3,581.53265	3,869.0398
R59	443.93226	492.18260	878.62399	611.84900	242.23098	550.29205	292.29479	502.15962	672.1034
R60	124.14491	123.71919	20.42639	67.83107	183.02301	101.70366	83.11590	135.51899	64.7014

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